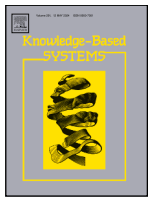
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Exploring non-negativity for improved manifold embedding: Application to t-SNE

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ABSTRACT

Drawing inspiration from Non-negative Matrix Factorization (NMF), this paper explores the potential of incorporating non-negativity constraints into embedding techniques, with a focus on t-SNE as an application. Specifically, we investigate the following questions: Can enforcing non-negativity in the embedding space enhance interpretability and improve the quality of embedded data? By prioritizing non-negativity, can embedding methods achieve better performance and more meaningful representations? Additionally, does enforcing non-negativity in the embedded space help preserve both the local and global structure of data in the manifold, leading to more accurate and interpretable embeddings? In this work, we could show both objectively and subjectively how enforcing t-SNE to leverage the non-negativity of the data addresses the raised questions. To achieve this, we introduced a novel approach to transforming the additive update rule of the gradient descent used by t-SNE to a multiplicative counterpart to enforce the non-negativity in the embedded space. However, grappling with full non-negativity in the gradient descent formula presents challenges, prompting our focus solely on the $(y_i - y_j)$ term, resulting in a semi-non-negative t-SNE algorithm, shortly named SN-tSNE. Nevertheless, experimental findings substantiate the significant impact of the proposed update rule on the performance and efficacy of the SN-tSNE algorithm. Furthermore, additional experiments are performed to compare SN-tSNE with its precursor t-SNE, as well as the competitive embedding technique UMAP, alongside other relevant embedding and dimensionality reduction models like NMF. The source code of SN-tSNE is available on GitHub (https://github.com/M-Allaoui/SN-tSNE.git).

1. Introduction

Non-negativity constraints have emerged as pivotal elements in dimensionality reduction and embedding techniques, showcasing numerous advantages across diverse domains, including image processing, text mining, and spectral data analysis. Non-negative Embedding (NNE) [1,2] operates as a dimensionality reduction technique, directly learning low-dimensional embeddings while upholding non-negativity constraints. NNE proves particularly advantageous in scenarios where the data points themselves are non-negative, such as word frequency matrices in text data. By preserving non-negativity, NNE ensures that resulting embeddings maintain the intrinsic structure and semantics of the original data, rendering them suitable for visualization [3], classification [4], and clustering tasks [5] across various domains, including hyperspectral image analysis [6] and document embedding for information retrieval [7]. Therefore, enforcing non-negativity constraints in dimensionality reduction and embedding techniques yields interpretable representations and induces sparsity, thereby enhancing computational efficiency and storage optimization, particularly in high-dimensional datasets. Among the non-negative techniques, Non-negative Matrix

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Factorization (NMF) [8,9] stands out as a widely employed method for feature extraction and dimensionality reduction, especially in contexts where the data inherently possesses non-negative attributes, such as images, text, and spectrograms. In NMF, a non-negative matrix $\mathbf{X}$ is decomposed into two non-negative matrices $\mathbf{W}$ (basis vectors) and $\mathbf{H}$ (encoding coefficients), i.e., $\mathbf{X} \approx \mathbf{WH}$, ensuring that both $\mathbf{W}$ and $\mathbf{H}$ solely contain non-negative elements. This property leads to parts-based representations, augmenting interpretability and facilitating tasks like feature extraction and pattern recognition in image processing [10,11] and document clustering in text mining [12]. However, subsequent research has suggested that relaxing the full non-negativity constraint to a semi-non-negativity setting can offer a balanced trade-off between interpretability and representational flexibility. Ding et al. [13] first introduced Semi-Nonnegative Matrix Factorization (SNMF), in which only one of the factor matrices (typically $\mathbf{W}$) is constrained to be non-negative while the other ($\mathbf{H}$) is allowed to take mixed signs. This relaxation maintains interpretability through the parts-based basis vectors while permitting richer representations and improved clustering quality. Semi-non-negativity has since become a common variant in applications where strict non-negativity is too restrictive, demonstrating its potential to balance structural fidelity and practical performance. It is also important to note that the benefits of non-negativity constraints are not universal. In some cases, such as computed tomography imaging, negative correlations between features are meaningful (e.g., financial data, gene expression data), unconstrained embeddings might be preferable where the flexibility of allowing negative coordinates can provide advantages [14]. The effectiveness of non-negative embeddings largely depends on the specific application, data characteristics, and the problem being addressed.

The exploration of integrating non-negativity constraints into manifold embedding techniques, particularly those like t-distribution Stochastic Neighbor Embedding (t-SNE) and Uniform Manifold Approximation and Projection (UMAP), which aim to preserve the local (and potentially global) structure of high-dimensional data, is a central motivation of this work. While non-negativity is inherently built into methods such as NMF, manifold learning techniques typically lack this constraint, potentially hindering the interpretability and semantic clarity of the resulting low-dimensional representations. By introducing non-negativity into manifold embeddings, we aim to improve both the structure and readability of the output space, particularly for applications in clustering, visualization, and reconstruction tasks. This approach aligns naturally with the goals of manifold learning, while addressing a known limitation in common methods.

t-SNE establishes relationships among data points in high-dimensional space and then seeks a low-dimensional representation that retains these relationships [15]. Achieving this involves defining probability distributions over pairs of data points in both high and low-dimensional spaces and minimizing the disparity between these distributions through gradient descent to optimize t-SNE's Kullback-Leibler (KL) divergence loss function. The resulting low-dimensional representation often unveils the underlying data structure, making it valuable for exploratory data analysis and visualization. However, t-SNE faces challenges, including limited scalability with substantial data volumes and a tendency to overlook global structure preservation. Criticisms also include its focus on local clusters [16,17]. In addition, traditional t-SNE embeddings may exhibit negative coordinates due to the additive nature of the gradient-descent updating rule, leading to challenges in interpretation and visualization. This limitation hampers the effectiveness of t-SNE in revealing underlying patterns in the data, where the non-negativity property is recommended. These challenges have prompted the exploration of alternative strategies for gradient descent updates to mitigate the negative embedded points, ultimately aimed at enhancing the performance, visualizability, and interpretability of t-SNE [18,19].

We chose t-SNE as the foundation for our proposed method due to its strong performance in capturing local structure and producing well-separated clusters, which are crucial for interpretable embeddings. While UMAP is known for its computational efficiency and better global structure preservation, several comparative studies, including the comprehensive benchmark by [20], demonstrate that t-SNE consistently ranks among the top-performing dimensionality reduction techniques in terms of average quality across diverse datasets. Specifically, t-SNE achieves higher scores on local structure metrics such as Trustworthiness, Neighborhood Hit, and Continuity, which are essential for tasks focused on clustering and neighborhood preservation. This empirical evidence supports our decision to enhance t-SNE rather than alternative methods. To this end, in this paper, we contribute to developing a novel approach to address these limitations by harnessing the power of enforcing non-negativity constraints on embedded points. We ensure that the resulting embedding contains fewer non-negative points, thereby enhancing interpretation and visualization by reducing the impact of negative embedded points.

Specifically, our work introduces a novel approach that transforms the additive gradient descent update rule in t-SNE into a multiplicative form to reduce negative values in the embedding space. Full non-negativity is difficult to impose due to the structure of the update rule; thus, we target only the $(y_i - y_j)$ term, resulting in a semi-non-negative variant, which we call SN-tSNE. This semi-non-negativity approach balances tractability with effectiveness, preserving the core structure of t-SNE's optimization while reducing the dominance of negative embeddings. In practice, it enables interpretability and improved convergence without requiring a full non-negative reparameterization, which may be computationally prohibitive or overly restrictive. By applying this Multiplicative Update Rule (MUR), SN-tSNE achieves superior performance compared to the original t-SNE. Additionally, MUR introduces a shrinkage effect on the embedding space, promoting tighter clusters and improved separation, critical for tasks like clustering and visualization. In summary, the contribution of the paper is:

1) Propose a new MUR to optimize the t-SNE's objective function, effectively reducing the number of negative points in the embedded space.
2) Through extensive experimentation, we objectively demonstrated that enforcing non-negativity in the embedding space enables the proposed algorithm (SN-tSNE) to consistently outperform the original t-SNE as well as other competitive embedding techniques across all synthetic and real datasets.
3) Using extensive synthetic and benchmark datasets, we demonstrated a positive correlation between the ratio of non-negative points in the embedding space and the quality of the embedding, as reflected in improved clustering performance. Thus, the fewer non-negative points in the embedding space, the better the clustering accuracy, indicating a more optimized data structure.
4) Leveraging the proposed MUR, the derived embedding technique achieves better results in fewer iterations, resulting in significantly reduced time complexity.
5) Subjectively, using face image datasets, we demonstrated that the proposed algorithm generates an embedding space that facilitates data reconstruction with lower error, resulting in more interpretable outputs.

2. Related work

Dimensionality reduction and manifold embedding techniques are essential tools for visualizing and analyzing high-dimensional data. Among these, NMF [8,9], UMAP, and t-SNE stand out as widely used methods. UMAP is particularly effective at preserving both local and global structures, but often sacrifices interpretability and compactness in clustering outcomes [21]. On the other hand, t-SNE excels at capturing local relationships within data, making it a popular choice for cluster

visualization. However, its reliance on additive update rules often leads to negative coordinates in the embedding space, which complicates interpretability. Furthermore, t-SNE faces challenges in scalability and fails to preserve global relationships crucial for comprehensive data analysis [15,17].

Despite its popularity, t-SNE has inherent limitations, particularly regarding interpretability and computational efficiency. Traditional t-SNE embeddings often include negative coordinates, which hinder the clarity of resultant visualizations. Additionally, its emphasis on local structure comes at the expense of global manifold relationships, limiting its applicability in domains requiring a broader perspective of the data [17]. To address these issues, researchers have proposed various extensions to t-SNE, such as modifying its loss function and optimization schemes [16,22,23]. Inspired by the concept of multiplicative updates used in NMF, several studies have explored incorporating these updates into t-SNE to improve convergence and performance. Yang et al. [24] were among the first to propose a multiplicative update rule for t-SNE, which we call MU-tSNE, enhancing its optimization process and stability. Similarly, van der Maaten et al. [25] investigated fast optimization techniques for t-SNE based on multiplicative updates, focusing on improving computational efficiency. Recent work by [19] provides a rigorous theoretical framework for analyzing the convergence properties of t-SNE. They show that under specific conditions on the pairwise affinity distributions, the optimization trajectory of t-SNE, driven by additive gradient descent, converges to a local minimum of the KL-divergence objective. However, they also highlight that convergence tends to be slow and sensitive to poorly conditioned or imbalanced affinities, which can hinder both efficiency and stability. Their analysis remains restricted to the additive optimization dynamics and does not extend to alternative formulations.

The discussion of all previous manifold embedding methods shows that they rarely addressed non-negativity constraints explicitly, leaving room for improvement in embedding interpretability. The integration of non-negativity constraints into dimensionality reduction techniques has proven highly beneficial, especially in domains such as image processing, text mining, and bioinformatics, where data inherently exhibit non-negativity. NMF is a foundational technique in this regard, decomposing data matrices into non-negative components to produce interpretable and sparse representations [2,8]. Advancements like graph-regularized NMF and deep non-negative factorization have further enhanced its utility in clustering and feature extraction tasks [1,3]. For instance, neighborhood-preserving NMF has been particularly effective in ensuring that learned factors retain local data structures. Gu et al. [26] integrated neighborhood information into the factorization process to preserve local relationships, and Mei et al. [27] extended this approach to spectral mixture analysis. Recent advancements have incorporated neighborhood structure directly into the learning process, as seen in [28], who proposed neighborhood-structure-assisted NMF for unsupervised anomaly detection, and Kaddouh et al. [29], who emphasized neighborhood preservation in embedding learning. These approaches underscore the significance of maintaining local relationships in low-dimensional representations. However, while NMF ensures interpretability through non-negativity, it struggles to model non-linear relationships effectively, a gap addressed by methods like t-SNE and UMAP.

Our proposed method, SN-tSNE, integrates the strengths of these existing works by combining multiplicative updates with non-negativity constraints and neighborhood preservation. While t-SNE captures the local structure of high-dimensional data effectively, its negative embeddings complicate interpretation. By enforcing non-negativity and leveraging a MUR inspired by prior research, SN-tSNE not only enhances the interpretability of embeddings but also preserves the neighborhood structure essential for meaningful data representation. This novel approach bridges the gap between interpretability and non-linear structure modeling, addressing limitations in existing dimensionality reduction techniques.

3. Background on t-SNE

t-SNE algorithm reduces the dimensions of a high D-dimensional data $X = \{x_1, x_2, \dots, x_n\}; x_i \in \mathbb{R}^D; \forall i$, to a lower d-dimensional space $Y = \{y_1, y_2, \dots, y_n\}; y_i \in \mathbb{R}^d; \forall i$, where $d \lll D$. The algorithm begins by computing pairwise similarities between data points in the high-dimensional space using the probability distribution p_{ij} that represents the likelihood of a point x_i choosing another point x_j as its neighbor from the data neighborhood set $\mathcal{N}_i^x$ of x_i. The probability distribution is computed using a Gaussian kernel, then it symmetrizes the similarities to obtain the joint probability distribution P defined as:

$$p_{j|i} = \frac{exp(-\left\|x_i - x_j\right\|^2/2\sigma_i^2)}{\sum_{k\neq l} exp(-\left\|x_k - x_l\right\|^2/2\sigma_i^2)}, p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n} \tag{1}$$

The Bandwidth parameter σ_i^2 in Eq. (1) is determined through binary search until a predefined perplexity τ value is achieved, where $\tau_{(i)} = 2^{\sum_j p_{j|i} \log_2 p_{j|i}}$. Subsequently, the Student's t-distribution with one degree of freedom is employed in the second step to calculate the similarities in the low-dimensional space. Each pair of low-dimensional points (y_i, y_j) is then represented as a probability distribution q_{ij}, defined as follows:

$$q_{ij} = \frac{(1 + \left\|y_i - y_j\right\|^2)^{-1}}{\sum_{k\neq l}(1 + \left\|y_k - y_l\right\|^2)^{-1}}$$

The objective of t-SNE is to discover the optimal low-dimensional representation Q, encompassing all low-dimensional pairwise probabilities q_{ij} that faithfully reflect P, consisting of all high-dimensional pairwise probabilities p_{ij}. In essence, the positions of the points y_i on the map are dictated by minimizing the Kullback-Leibler divergence (KL) between the distribution P and the distribution Q. This divergence, being non-symmetric, is defined as:

$$C = KL(P\|Q) = \sum_i KL(P_i\|Q_i) = \sum_i \sum_{j\neq i} p_{ij} \log \frac{p_{ij}}{q_{ij}}.$$

The KL divergence is minimized with respect to y_i using the Gradient Descent update rule.

$$y_i^{t+1} = y_i^t + \eta \frac{\delta C}{\delta y_i} + \alpha(t)(y_i^{t-1} - y_i^{t-2}), \tag{2}$$

With the gradient defined by

$$\frac{\delta C}{\delta y_i} = 4 \sum_j (p_{ij} - q_{ij})(y_i - y_j)(1 + \left\|y_i - y_j\right\|^2)^{-1}, \tag{3}$$

where η denotes the learning rate, and $\alpha(t)$ represents the momentum at iteration t. The learning rate is a crucial factor that significantly influences model performance. Instead of relying on a fixed learning rate throughout the training process, in the original t-SNE, the learning rate is updated increasingly according to the description of Jacobs [22] to speed up the performance of the models. The momentum term controls the influence of previous updates on the current update direction, which aids in accelerating convergence by reducing the oscillations in the optimization process. Specifically, t-SNE incorporates a momentum term that dynamically adjusts based on the iteration count. Initially, a smaller momentum term is preferred when the map points are poorly organized. For instance, the momentum term was set to $\alpha(t) = 0.5$ for $t < 250$ and $\alpha(t) = 0.8$ for $t \geq 250$, allowing for smoother transitions between iterations and facilitating effective optimization.

4. Proposed method

Our work is motivated by the success of NMF and its MUR. It aims to leverage the power of no-negativity in a novel t-SNE embedding framework to provide a natural and intuitive way to interpret data. This section will present the mathematical foundation of integrating non-negativity into the t-SNE framework via two distinct strategies. The first strategy involves transforming the additive gradient descent update rule

into a multiplicative counterpart, akin to the optimization principle employed in the NMF approach. In contrast, the second strategy is a simple application of Projected Gradient Descent (PGD) after each gradient descent update to ensure that the solution remains non-negative during optimization. However, we can claim that the contribution of our work lies in the first strategy because it exerts a more significant influence on the quality of the results. Therefore, we consider the PGD-based update rule as an integral part of our experimental evaluation and comparison, serving to highlight the impact of the proposed MUR.

4.1. SN-tSNE

Semi-non-negative t-SNE Eq. (2), which represents an additive update rule of the y-coordinates, involves three terms that may generate negatives values, $(p_{ij} - q_{ij})$, $(y_i - y_j)$ and $(y_i^{t-1} - y_i^{t-2})$. In this work and for the easiest formulation of the proposed update rule of y-coordinates, we set $\alpha(t) = 0; \forall t$, which effectively cancels out the effect of the momentum term, resulting in the following simplified rule:

$$y^{t+1} = y^t + \eta \frac{\delta C}{\delta y}, \text{ for all } y \in Y, \tag{4}$$

with

$$\frac{\delta C}{\delta y} = 4 \sum_j T_j (y - y_j), \tag{5}$$

such that $T_j = (p_{ij} - q_{ij})(1 + \left\| y - y_j \right\|^2)^{-1}$ and $y = y_i$ (i.e., the subscript i is omitted for clarity). Therefore, to mitigate the presence of non-negative values, we suggest transitioning from an additive rule of the gradient descent to update the y-coordinates to a multiplicative rule as defined in the following Eqs. (6) or (7). Hence, if the y-coordinates are initialized with positive values, they will remain positive, provided that $(p_j - q_j)$ is assumed to be positive. However, as the latter condition is not always fulfilled, which can be either positive or negative, depending on mismatches in affinities between high- and low-dimensional spaces, leading to what we term semi-non-negativity. Directly constraining the full gradient to be non-negative could hinder descent directions and destabilize training. Our semi-non-negative formulation circumvents this by focusing on directional movement, which retains the optimization behavior of t-SNE while enhancing convergence and interpretability. Exploring other strategies to achieve full non-negativity, such as projected gradients (as discussed in Section 4.3) or reparameterization, could enforce strict non-negativity. They may fundamentally alter the geometry and optimization dynamics of the original t-SNE algorithm, potentially compromising its capacity to faithfully preserve pairwise affinities and local structure. With this aim in mind, we present the following theorem:

Theorem 1. *Let* υ *denote the vector* $\left[\left(\sum_j (T_j y_{j1}) + \epsilon\right)^{-1}, \left(\sum_j (T_j y_{j2}) + \epsilon\right)^{-1}, \ldots, \left(\sum_j (T_j y_{jd}) + \epsilon\right)^{-1}\right]^\top$*; where* y_{jk} *is the* k^{th} *element of the vector* y_j*. By setting an adaptive learning rate* η *to* $\frac{1}{4} y \odot \upsilon$*, the additive update rule in Eq. (4) can be transformed to a MUR of the following form (the superscript* t *is omitted for clarity):*

$$y = y \odot y \odot \upsilon \times \left(\sum_j T_j \right), \tag{6}$$

where $\odot$ *is the element-wise multiplication. The problem defined by Eq. (6) is solved element-wise, i.e, for each element* k *of the vector* y*, as follows:*

$$y_k = y_k^2 \times \left(\sum_j T_j y_{jk} + \epsilon \right)^{-1} \times \left(\sum_j T_j \right); k = 1.d. \tag{7}$$

Proof. By substituting the gradient in (4) by Eq. (5), we have:

$$y = y + 4\eta \sum_j T_j (y - y_j). \tag{8}$$

Since the derivation will be performed for each element k of the vector y and by adapting the learning rate η_k to each k, $k = 1.d$, we can obtain the following:

$$\begin{aligned}
y_k &= y_k + 4\eta_k \sum_j T_j (y_k - y_{jk}), \\
&= y_k + 4\eta_k \Big(T_1 y_k - T_1 y_{1k} + T_2 y_k - T_2 y_{2k} \\
&\qquad + \cdots + T_n y_k - T_n y_{nk} \Big), \\
&= y_k + 4\eta_k \Big((T_1 y_k + T_2 y_k + \cdots + T_n y_k) \\
&\qquad - (T_1 y_{1k} + T_2 y_{2k} + \cdots + T_n y_{nk}) \Big), \\
&= y_k + 4\eta_k \sum_j T_j y_k - 4\eta_k \sum_j T_j y_{jk}, \\
&= y_k + 4\eta_k y_k \sum_j T_j - 4\eta_k \sum_j T_j y_{jk}.
\end{aligned} \tag{9}$$

By setting η_k to $\frac{y_k}{4 \sum_j T_j y_{jk}}$, we obtain the following equation:

$$\begin{aligned}
y_k &= y_k + 4 \frac{y_k}{4 \sum_j T_j y_{jk}} y_k \sum_j T_j - 4 \frac{y_k}{4 \sum_j T_j y_{jk}} \sum_j T_j y_{jk}, \\
&= y_k + \frac{y_k \times y_k \times \sum_j T_j}{\sum_j T_j y_{jk}} - y_k, \\
&= y_k \times y_k \times \left(\sum_j (T_j y_{jk}) + \epsilon \right)^{-1} \times \left(\sum_j T_j \right).
\end{aligned} \tag{10}$$

Therefore, since $\upsilon = \left[\left(\sum_j (T_j y_{j1}) + \epsilon\right)^{-1}, \left(\sum_j (T_j y_{j2}) + \epsilon\right)^{-1}, \ldots, \left(\sum_j (T_j y_{jd}) + \epsilon\right)^{-1}\right]^\top$, the vectorial form of y can be written as follows:

$$y = y \odot y \odot \upsilon \times \left(\sum_j T_j \right).$$

□

Eq. (6) defines a multiplicative update rule that enforces non-negativity on the directional term $(y_i - y_j)$ and induces a natural *shrinkage effect* through the $y_k \times y_k$ term. This shrinkage compresses distances between points within clusters while maintaining separation between distinct clusters, thus enhancing compactness and convergence speed. To provide an intuitive understanding of the shrinkage effect introduced by the MUR, we consider a simplified 2D example involving three points initialized in a triangular configuration around an implicit centroid $[1.0, 1.0]$. Let the initial embeddings be:

$$y_1 = [0.9, 1.0], \quad y_2 = [1.1, 0.95], \quad y_3 = [1.0, 1.05]$$

Assume for illustration that the T_j weights are constant and normalized (e.g., $T_j = 1/3$ for all j), and the update is computed using Eq. (7). After one iteration, we get:

$$y_{ik}^{\text{new}} = y_{ik}^2 \cdot \frac{\sum_j T_j}{\sum_j T_j y_{jk}}, \quad \text{for each } i = 1, 2, 3 \text{ and } k = 1, 2$$

Applying this yields:

$$y_1^{\text{new}} \approx [0.89, 0.99], \quad y_2^{\text{new}} \approx [1.09, 0.94], \quad y_3^{\text{new}} \approx [0.99, 1.04]$$

We observe a consistent contraction toward the cluster centroid while maintaining relative positioning, which validates the expected shrinkage effect of MUR. Repeating the update over multiple iterations leads to increasingly compact and stable clusters.

4.2. Convergence theory: Additive vs. multiplicative updates

To better understand the theoretical distinction between the classical t-SNE optimization and our proposed SN-tSNE, we explicitly compare their update dynamics. Standard t-SNE employs an additive gradient descent rule as in Eq. (4). The additive formulation allows each

update step to move in directions dictated by both attractive and repulsive forces, but it does not prevent the embedding coordinates from taking negative values. Moreover, the sign ambiguity in $(p_{ij} - q_{ij})$ may lead to oscillations or slow convergence in certain regions of the optimization landscape. In contrast, our semi-non-negative formulation transforms this additive update into a multiplicative update rule as in Eq. (7). This multiplicative form introduces two key theoretical distinctions: 1) **Semi-Non-Negativity:** If the initialization is non-negative, the multiplicative structure ensures that updates remain predominantly non-negative, except for sign variations inherited from $(p_{ij} - q_{ij})$. This prevents embeddings from drifting into heavily negative regions, improving interpretability. 2) **Shrinkage Effect:** The quadratic term y_k^2 in Eq. (7) induces a contraction around local centroids, tightening clusters while preserving neighborhood relations. This modifies the convergence dynamics compared to the additive case, often leading to faster stabilization of the embedding. Thus, while classical t-SNE relies solely on additive adjustments that can oscillate around equilibrium, SN-tSNE's multiplicative rule provides directional bias toward stable, compact configurations. This modification preserves t-SNE's local structure fidelity while improving convergence behavior and interpretability.

4.3. PGD-tSNE: t-SNE with projected gradient descent optimization

Projected gradient descent is a method used to apply constraints on optimization variables during the optimization process. When dealing with optimization problems where the solution must satisfy certain constraints, such as non-negativity, projected gradient descent ensures that each update step stays within the feasible region defined by these constraints. Implementing t-SNE with projected gradient descent optimization involves modifying the standard t-SNE algorithm to incorporate the projection step to maintain the non-negativity constraint. After each update step in Eq. (4), we project the embedding coordinates onto the non-negative space to enforce the non-negativity constraint.

$$y_i^{(t+1)} = \max(y_i^t + \eta \frac{\delta C}{\delta y_i}, 0); \forall i, k \tag{11}$$

5. Experiments: Evaluation and validation

This section evaluates the performance of SN-tSNE in terms of the clustering task compared to t-SNE [15], PGD-tSNE, MU-tSNE [24], UMAP [21], and NMF [8,9] on benchmark datasets using various evaluation metrics detailed in Section 5.1. The analysis of perplexity and the number of iterations settings for SN-tSNE and t-SNE are given in Section 5.2. Section 5.3 investigates the influence of the Non-Negative Points Ratio (NNP Ratio) on the quality of the embedding space. The computational complexity and the convergence are addressed in Sections 5.4 and 5.5. The shrinkage effect of MUR on inter-Point distances is studied in Section 5.6 and the structure preservation is addressed in Section 5.7. In addition, an extensive evaluation of SN-tSNE's performance in terms of clustering, classification, and visualization tasks compared to the baseline techniques is provided in Sections 5.8 and 5.9. Finally, the interpretability aspect is studied in Section 5.10.

5.1. Datasets and evaluation metrics

Datasets. Eight well-known datasets are used in the conducted experiments, namely, Pendigits, USPS, MNIST-Digit, Fashion MNIST, Olivetti Faces, COIL-20, Isolet, and UCI-HAR. These datasets are chosen due to the positivity of their values since they contain grayscale images. **Fashion-MNIST** (F-MNIST) [30] comprises 70,000 samples distributed across 10 classes, with each sample being a 28×28 image depicting various fashion items. **MNIST** consists of 70,000 handwritten digits divided into 10 classes, with each image being a 28×28 grayscale image [31]. The **Olivetti Faces** dataset contains 400 grayscale images of 40 individuals, with 10 grayscale images of size 64×64 pixels per person, capturing variations in expressions, lighting, and facial details. The **USPS** dataset is a collection of digit images automatically scanned from envelopes by the U.S. Postal Service, containing a total of 9,298 16×16 pixel grayscale samples. These images are centered, normalized, and display a diverse array of font styles [32]. **Pendigits** [33] comprises 1797 images representing 10 classes of numbers from 0 to 9, each with a size of 8×8 pixels. The Columbia Object Image Library **COIL-20** dataset consists of 1440 grayscale images with size 128×128 pixels of 20 objects, each captured at 72 different angles (rotated in 5-degree increments). **Isolet** [34] is a speech-based dataset with 7,797 samples representing spoken letters (A-Z) from 150 speakers, each described by 617 acoustic features. The task is to classify each sample into one of 26 spoken letter categories, providing a high-dimensional, non-image benchmark for evaluating the applicability of manifold learning techniques to auditory domains. **UCI Human Activity Recognition (UCI-HAR)** dataset [35] contains 10,299 samples of multivariate time-series sensor data collected from smartphones. Each instance is represented by 561 extracted features and labeled into one of 6 human activities (e.g., walking, sitting, lying). The dataset provides a non-visual, continuous sensor modality ideal for testing generalization beyond image data. The key characteristics of all the mentioned datasets are presented concisely in Table 1.

Table 1
Key characteristics of benchmark datasets.

Dataset	# of samples	# of features	# of labels
FMNIST	70000	784	10
MNIST	70000	784	10
Olivetti Faces	400	4,096	40
USPS	9298	256	10
Pendigits	1797	64	10
COIL20	1440	16384	20
Isolet	7797	617	26
UCI-HAR	10299	561	6

Evaluation Metrics. We use Shapiro-Wilk (SW) and Kolmogorov-Smirnov (KS) tests to evaluate the convergence performance, where the larger the basis p-value, indicates better the performance. In addition, we assess the performance of the K-means clustering algorithm across varied embedding spaces derived from the aforementioned techniques using performance measures sourced from the Scikit-learn Python library [1], Accuracy (ACC), Normalized Mutual Information (NMI), Silhouette coefficient (SC), and Davis-Bouldin Index (DBI). For ACC and NMI, the best performance should be close to 1, while the poor performance is near 0. Higher SC values indicate a better clustering structure, while -1 suggests suboptimal clustering. The DBI varies from 0 to infinity, with lower values signifying superior clustering quality. To address the structure preservation aspect, we used three established metrics, which are Global Quality (GQ) [37], Mean Relative Rank Error (MRRE) [38], and Trustworthiness [39].

5.2. Parameter tuning

To ensure fair and optimal comparisons, we tuned the key parameters of SN-tSNE and t-SNE, perplexity, and number of iterations, across all datasets. These parameters significantly influence convergence behavior and embedding quality, cluster separation, stability, and computational cost.

Perplexity. This part compares SN-tSNE and the original t-SNE (baseline) across different values of Perplexity (τ). Both techniques' Perplexity (τ) parameter is systematically adjusted within the [5, 50] range using all datasets. Table 2 reveals the accuracy of the k-means clustering algorithm for both algorithms, where the best accuracy score is selected

[1] https://scikit-learn.org/stable/modules/classes.html#module-sklearn.metrics

Table 2
Accuracy of SN-tSNE and t-SNE with various Perplexity values based on all datasets.

Datasets	Algorithm	Perplexity (τ)						
		5	10	15	20	25	35	45
F-MNIST	t-SNE	0.53 ± 0.04	0.53 ± 0.02	0.54 ± 0.02	0.55 ± 0.03	0.56 ± 0.02	0.57 ± 0.01	0.55 ± 0.02
	SN-tSNE	0.55 ± 0.02	0.56 ± 0.02	0.57 ± 0.01	**0.58 ± 0.02**	0.57 ± 0.02	0.56 ± 0.01	0.55 ± 0.01
MNIST	t-SNE	0.78 ± 0.02	0.79 ± 0.03	0.79 ± 0.02	0.80 ± 0.03	0.81 ± 0.02	0.82 ± 0.02	0.81 ± 0.02
	SN-tSNE	0.82 ± 0.02	0.83 ± 0.02	0.84 ± 0.02	**0.85 ± 0.02**	0.84 ± 0.02	0.83 ± 0.02	0.81 ± 0.02
Olivetti	t-SNE	0.64 ± 0.01	0.65 ± 0.01	0.64± 0.01	0.63± 0.01	0.61± 0.01	0.60± 0.01	0.58± 0.01
	SN-tSNE	**0.67 ± 0.01**	0.66 ± 0.01	0.66± 0.01	0.64± 0.01	0.63± 0.01	0.62± 0.01	0.60± 0.01
USPS	t-SNE	0.76 ± 0.02	0.77 ± 0.03	0.77 ± 0.04	0.79 ± 0.02	0.80 ± 0.02	0.79 ± 0.03	0.78 ± 0.04
	SN-tSNE	0.82 ± 0.02	**0.83 ± 0.02**	0.82 ± 0.03	0.81 ± 0.03	0.80 ± 0.02	0.79 ± 0.02	0.78 ± 0.02
Pendigits	t-SNE	0.89 ± 0.02	0.85 ± 0.04	0.82 ± 0.06	0.81 ± 0.06	0.79 ± 0.06	0.77 ± 0.06	0.75 ± 0.06
	SN-tSNE	**0.90 ± 0.02**	0.89 ± 0.02	0.87 ± 0.04	0.86 ± 0.04	0.86 ± 0.04	0.83 ± 0.04	0.81 ± 0.04
COIL-10	t-SNE	0.38± 0.04	0.57± 0.01	0.77± 0.01	0.83± 0.01	0.79± 0.01	0.74± 0.01	0.71± 0.01
	SN-tSNE	0.85± 0.01	0.87± 0.01	**0.88± 0.01**	0.86± 0.01	0.84± 0.01	0.83± 0.01	0.82± 0.01
Isolet	t-SNE	0.56±0.01	0.58±0.01	0.60±0.01	0.59±0.01	0.59±0.01	0.58±0.01	0.57±0.01
	SN-tSNE	**0.63±0.01**	0.62±0.01	0.62±0.01	0.61±0.01	0.61±0.01	0.60±0.01	0.59±0.01
UCI-HAR	t-SNE	0.64±0.01	0.61±0.01	0.62±0.01	0.65±0.01	0.64±0.01	0.63±0.01	0.61±0.01
	SN-tSNE	0.66±0.01	0.69±0.01	**0.70±0.01**	0.69±0.01	0.68±0.01	0.67±0.01	0.65±0.01

over all the scores of other perplexities in each dataset. The perplexity values where both algorithms achieved their best scores are used in the performed experiments. From the table, it is evident that, for most of the datasets, SN-tSNE performs better with smaller perplexity values. This may imply that SN-tSNE places greater emphasis on preserving local similarities among neighboring data points within the high-dimensional space, especially considering that the datasets lack well-defined global structures. In future research, a comprehensive analysis of the relationship between perplexity and cluster structure can be carried out by integrating various types of data distributions through a synthetic generation process, potentially as a separate study.

Number of Iteration. This experiment analyzes the number of iterations required by both t-SNE and SN-tSNE algorithms. Figs. 1 and 2 illustrate that SN-tSNE converges to the optimal embedding state, reflected by the highest clustering performance (Accuracy and NMI), with fewer iterations (∼ 50), whereas t-SNE requires approximately 200 iterations on the Pendigits dataset. Regarding the Isolet, UCI-HAR, USPS, Olivetti, COIL-20, MNIST, and F-MNIST datasets, SN-tSNE requires respectively ∼ 100, ∼ 100, ∼ 100, ∼ 150, ∼ 150, ∼ 250, and ∼ 250 to converge, while t-SNE needs ∼ 500, ∼ 500, ∼ 500, ∼ 500, ∼ 500, ∼ 1000, and ∼ 1000 iterations to achieve its best scores.

5.3. Impact of non-negativity

To investigate the influence of the Non-Negative Points Ratio (NNP Ratio) on embedding quality, particularly clustering performance measured by NMI and ACC, we evaluated SN-tSNE on both controlled synthetic datasets and real-world benchmark datasets. This evaluation allows us to highlight the algorithm's behavior in well-defined settings while also demonstrating its robustness under practical scenarios.

Synthetic dataset. We first conducted experiments on synthetic Blob datasets, which were carefully configured with varying numbers of features and cluster centers to provide a controlled test environment. These datasets allow us to systematically analyze how SN-tSNE responds to increasing complexity. The correlation between clustering metrics and the NNP ratio is presented in Table 3 and visualized in Fig. 3(a) and (b). Results consistently revealed a strong positive correlation (NMI = 0.99, ACC = 0.99), indicating that as the NNP ratio increases, clustering performance improves significantly. This confirms the role of non-negativity in promoting clearer cluster separation in simplified, noise-free settings.

Table 3
The impact of the non-negative points (NNP) ratio on the performance of SN-tSNE on the Blobs (synthetic) datasets.

Blobs Dataset	# of Centers	# of Features	NNP Ratio	NMI	ACC
1	3	100	0.33	0.38	0.40
2	6	80	0.35	0.33	0.34
3	6	10	0.46	0.43	0.45
4	10	20	0.49	0.51	0.53
5	3	80	0.50	0.50	0.51
6	6	20	0.50	0.52	0.54
7	3	10	0.51	0.52	0.54
8	10	10	0.52	0.52	0.55
9	3	20	0.52	0.53	0.56
10	10	100	0.54	0.58	0.59
11	6	100	0.58	0.60	0.62
12	3	40	0.64	0.68	0.70
13	10	40	0.67	0.68	0.71
14	10	80	0.76	0.80	0.81
15	6	40	0.83	0.87	0.89

Correlation between NNP Ratio and NMI = 0.99, ACC = 0.99

Real-world benchmark datasets. To validate these findings beyond synthetic scenarios, we extended our analysis to benchmark datasets such as MNIST, F-MNIST, Olivetti Faces, USPS, Isolet, Pendigits, UCI-HAR, and COIL-20. Here, SN-tSNE was compared against classical t-SNE to assess the added value of enforcing non-negativity. The results, reported in Table 4 and illustrated in Fig. 4(a) and (b), demonstrate consistent performance gains for SN-tSNE across datasets. For instance, Olivetti Faces showed an NMI increase from 0.83 to 0.92, while COIL-20 improved from 0.94 to 0.96. Correlation analysis further revealed stronger alignment between NNP ratio and clustering quality for SN-tSNE (NMI AP = 0.40, ACC AP = 0.50) compared to t-SNE. These findings highlight that non-negativity not only enhances clustering accuracy but also improves interpretability in realistic, high-dimensional data. The combined evidence from synthetic and real-world experiments substantiates the critical role of non-negativity in improving the stability, clarity, and interpretability of manifold embeddings, both in controlled and practical settings.

5.4. Convergence analysis

In the convergence analysis, we employed the SW and KS tests to assess the normality of the error distributions for both the SN-tSNE and t-SNE algorithms using p-values as the basis for rejecting the hypothesis

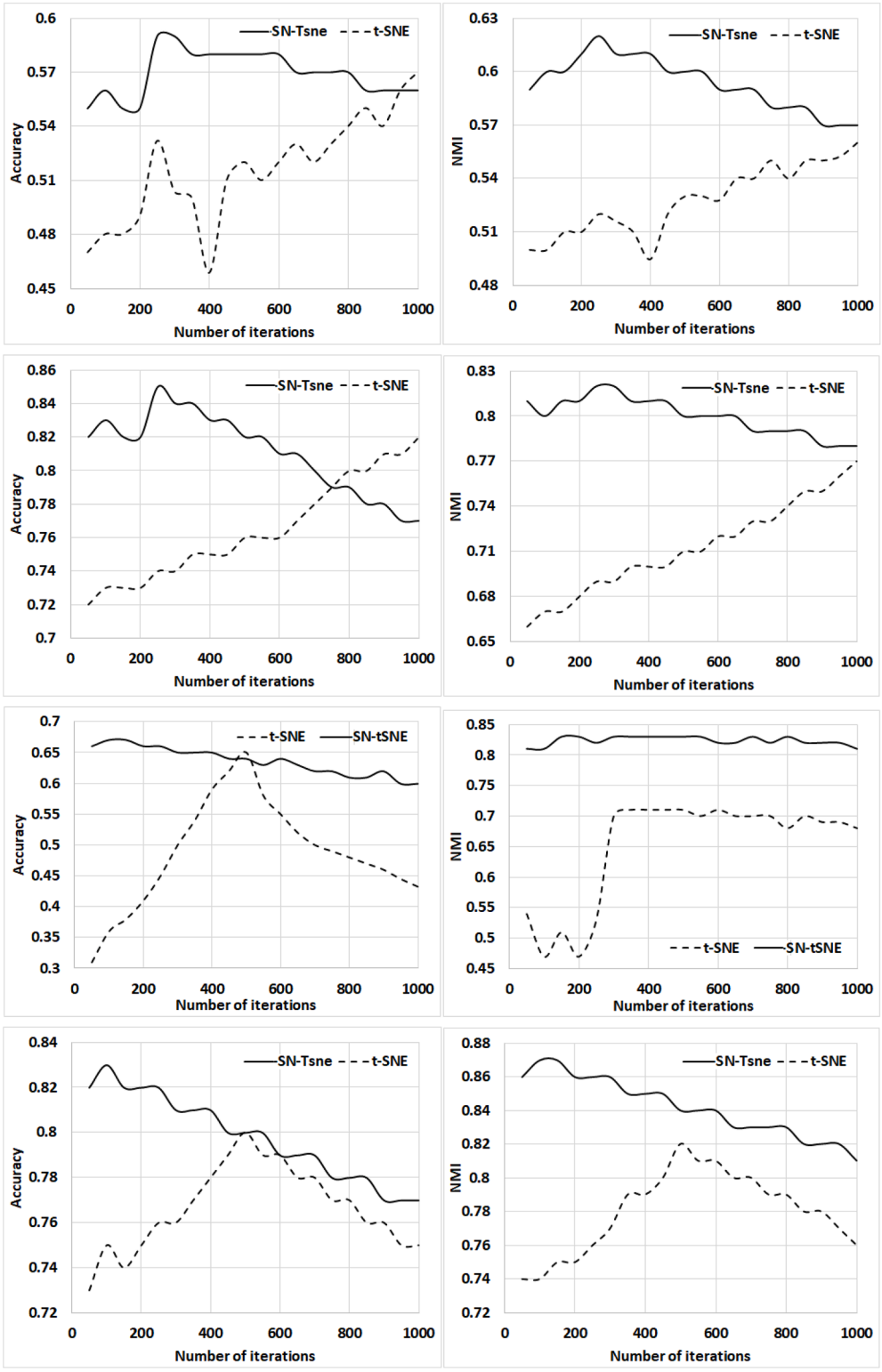


Fig. 1. The performance of t-SNE (dashed lines) and SN-tSNE (solid lines) measured by Accuracy and NMI over iterations, for the F-MNIST, MNIST, Olivetti faces, and USPS datasets.

of normality. If the p-value is smaller, it indicates a significant deviation from normality, while larger p-values suggest that the errors follow a normal distribution. To conduct this analysis, we ran each algorithm 50 times, recording the loss values (errors) generated during each run. We then analyzed the respective error distributions to determine whether they approximated a normal distribution. As a hypothesis, we assumed that a more normal error distribution would indicate more stable and predictable convergence behavior. Table 5 reports the p-values for each test analysis. From the results, it is clear that the p-value obtained from each test for SN-tSNE is greater than that for t-SNE. This suggests that SN-tSNE exhibits a better fit to the normal distribution, implying that its error distribution is closer to normality compared to t-SNE. Since

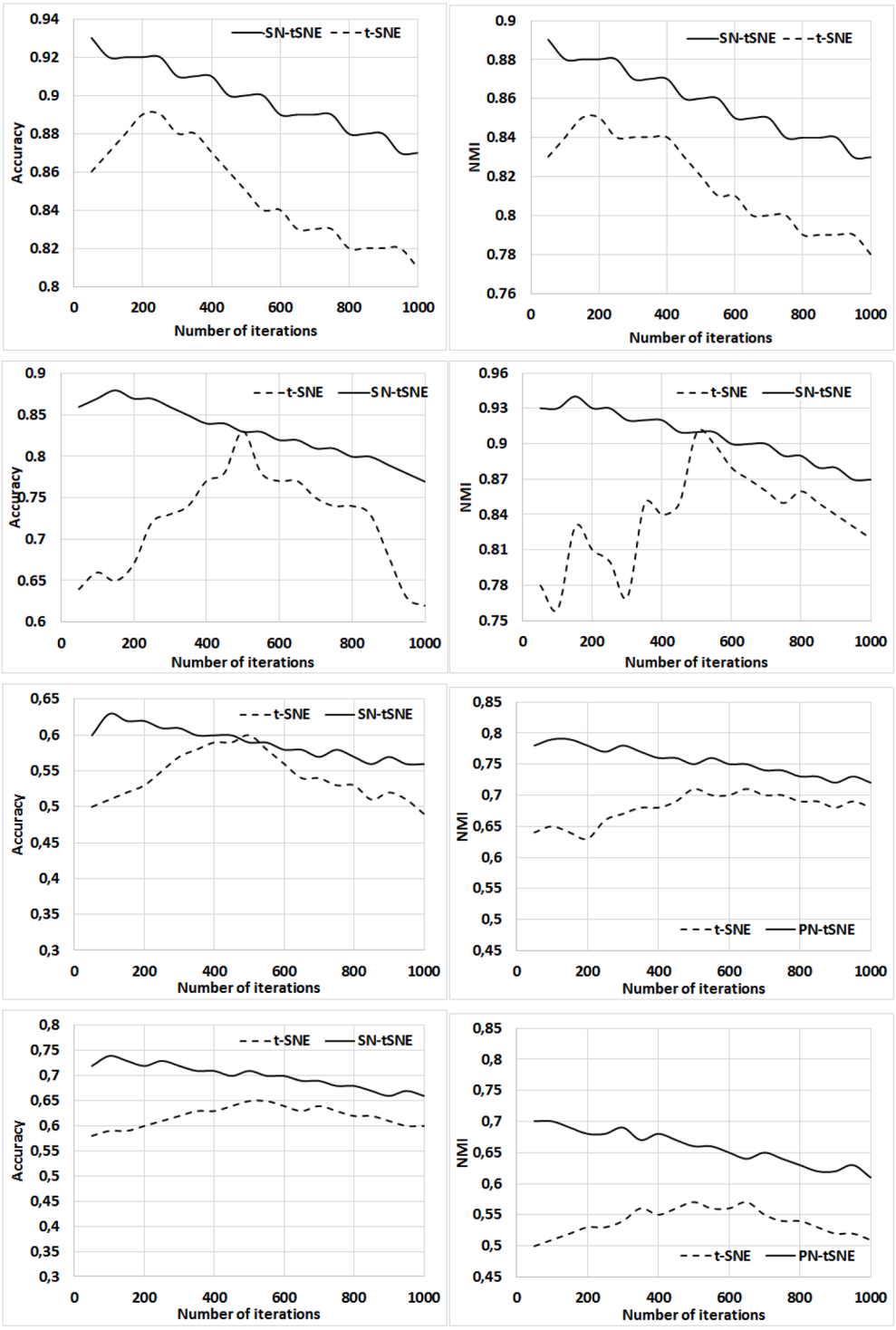


Fig. 2. The performance of t-SNE (dashed lines) and SN-tSNE (solid lines) measured by Accuracy and NMI over iterations for Pendigits, COIL-20, Isolet, and UCI-HAR datasets.

normality is often associated with more stable and predictable behavior in iterative algorithms, these results support the idea that SN-tSNE converges more reliably than t-SNE.

Fig. 5(a)–(h) present Kernel Density Estimation (KDE) and Quantile-Quantile (Q-Q) plots comparing SN-tSNE and t-SNE on the benchmark datasets. These visualizations offer subjective insights and a visual analysis of the normality of the loss distributions. The KDE plots show SN-tSNE (blue) with a well-formed bell-shaped curve that is less fat and closer to a normal distribution in most cases, indicating better alignment with normality. In contrast, t-SNE (red) exhibits mild skewness and a broader distribution. In the Q-Q plots, SN-tSNE better aligns with the diagonal, confirming a good fit to normality, while t-SNE shows a little deviation, particularly in the tails.

5.5. Computational complexity analysis

This section addresses the execution time of SN-tSNE against other embedding techniques on all datasets. Table 6 provides the execution

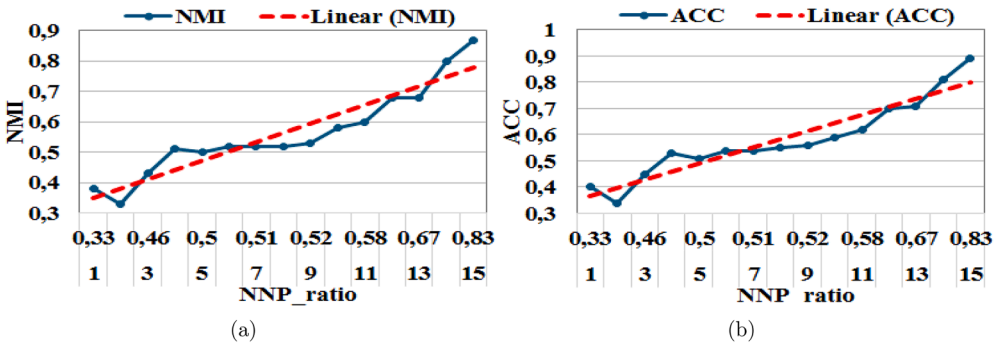


Fig. 3. Impact of the Non-Negative Points (NNP) ratio on clustering performance for the Blobs dataset using clustering metrics NMI (left) and Accuracy (right). The blue line shows clustering metrics results as the NNP ratio increases, with the dashed red line highlighting the positive relationship between the NNP ratio and clustering metrics.

Table 4
The impact of the non-negative points (NNP) ratio on embedding using the benchmark datasets.

Dataset	Method	NMI AP	NMI NNP	ACC AP	ACC NNP	NNP Ratio
F-MNIST	t-SNE	0.56	0.54	0.57	0.58	0.41
	SN-tSNE	0.62	0.66	0.58	0.60	0.50
MNIST	t-SNE	0.77	0.76	0.82	0.80	0.41
	SN-tSNE	0.82	0.85	0.85	0.88	0.51
Olivetti	t-SNE	0.82	0.81	0.65	0.63	0.43
	SN-tSNE	0.83	0.92	0.67	0.70	0.51
USPS	t-SNE	0.82	0.79	0.80	0.81	0.43
	SN-tSNE	0.87	0.89	0.83	0.85	0.53
Isolet	t-SNE	0.70	0.69	0.60	0.61	0.48
	SN-tSNE	0.79	0.80	0.63	0.65	0.54
Pendigits	t-SNE	0.86	0.83	0.89	0.87	0.47
	SN-tSNE	0.89	0.93	0.92	0.93	0.56
UCI-HAR	t-SNE	0.57	0.58	0.65	0.66	0.49
	SN-tSNE	0.70	0.72	0.74	0.77	0.58
COIL-20	t-SNE	0.91	0.90	0.83	0.82	0.50
	SN-tSNE	0.94	0.96	0.88	0.90	0.59
	t-SNE	**Correlation between NNP Ratio and NMI AP = 0.16, ACC AP = 0.13**				
	SN-tSNE	**Correlation between NNP Ratio and NMI AP = 0.40, ACC AP = 0.50**				

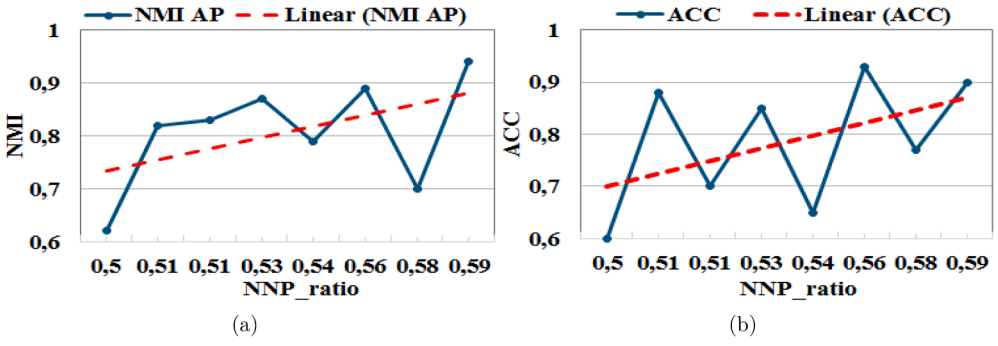


Fig. 4. Impact of the Non-Negative Points (NNP) ratio on clustering performance on benchmark datasets using clustering metrics NMI (left) and Accuracy (right). The blue line shows clustering metrics results as the NNP ratio increases, with the dashed red line highlighting the positive relationship between the NNP ratio and clustering metrics.

times (in seconds) for all methods under identical computational conditions. The results demonstrate that SN-tSNE is the second-fastest algorithm behind NMF. NMF is generally considered a straightforward technique based on matrix algebra; unlike other techniques, it doesn't involve computing probabilities, contributing to its speed. It is clear that UMAP outperforms t-SNE in three datasets. UMAP outpaces MU-tSNE in terms of speed on MNIST and FMNIST, while MU-tSNE is much faster than the other techniques on Pendigit and USPS datasets. The PDG-tSNE trails in execution time, requiring a substantial number of iterations to optimize its performance. These results underscore the efficiency of SN-tSNE in handling large datasets, which is crucial for applications requiring rapid dimensionality reduction and embedding generation. The significant reduction in execution time without compromising embedding quality demonstrates the practicality of our method in real-world scenarios.

Additionally, we analyzed the time complexity of the SN-tSNE algorithm and others with respect to data size using the USPS dataset as a case study. This is shown in Fig. 6. The results demonstrate that

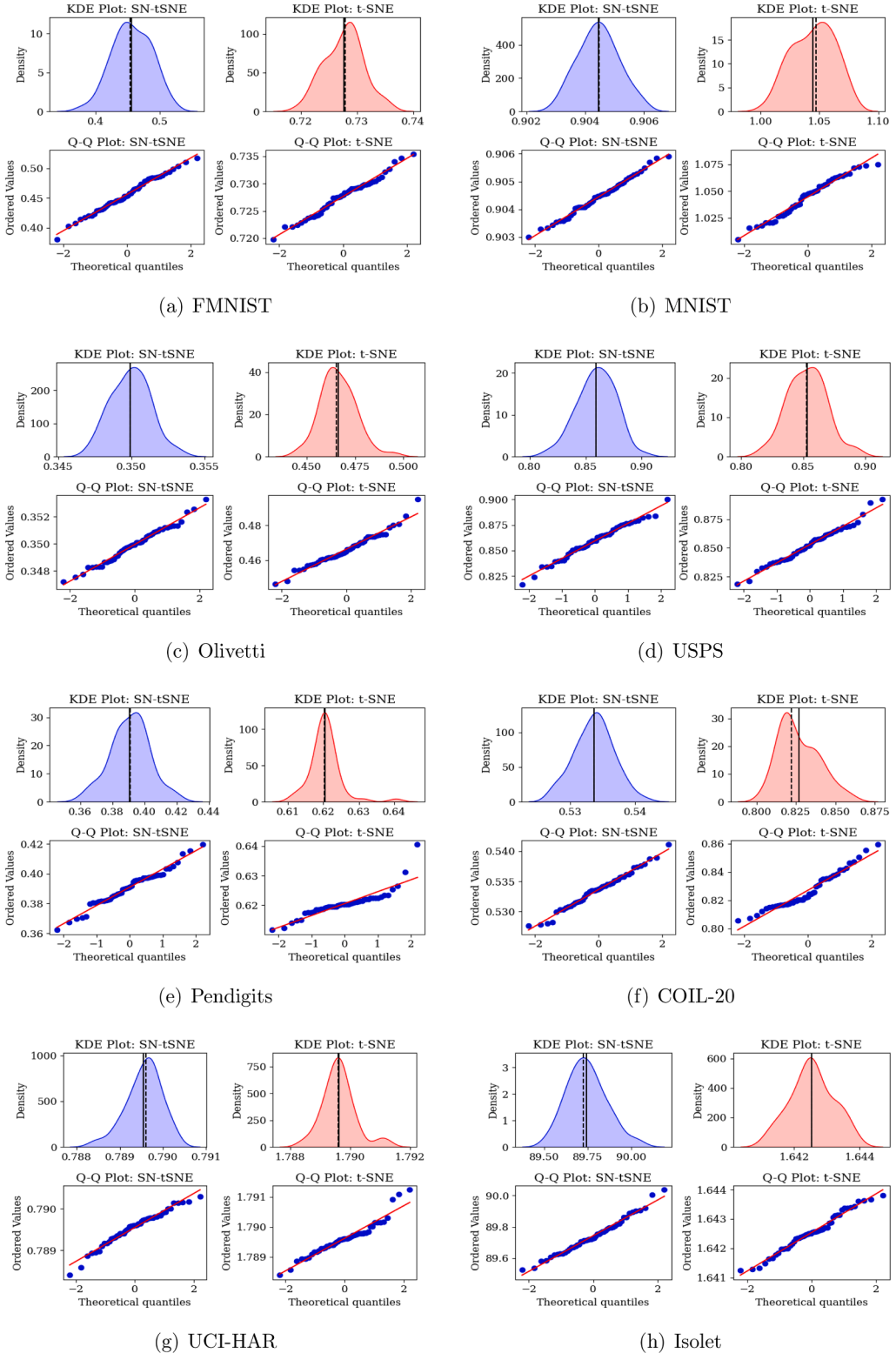


Fig. 5. Visual analysis of the normality of SN-tSNE and t-SNE loss distributions using KDE and Q-Q plots.

SN-tSNE exhibits superior scalability, maintaining the lowest execution time across all dataset sizes. This efficiency highlights the optimization benefits of our approach, where using the multiplicative updates rule to enforce non-negativity constraints reduces computational overhead compared to t-SNE and its variants. Notably, standard t-SNE shows significantly higher execution times as the data size exceeds 5000 samples, due to its reliance on computationally intensive additive update rules.

Similarly, PGD-tSNE and MU-tSNE, while improved over standard t-SNE, show increased execution times as the dataset grows, reflecting their more complex optimization procedures. UMAP, known for its

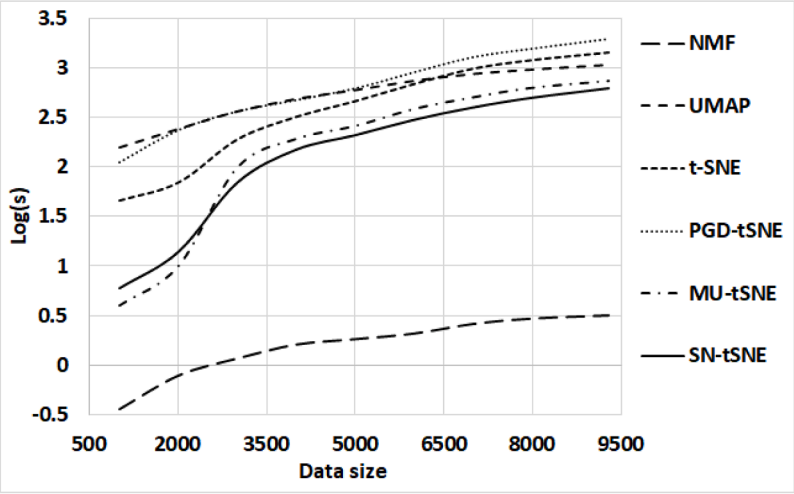


Fig. 6. Log-scale execution time comparison of SN-tSNE and baselines on the USPS dataset as data size increases.

Table 5
Normality analysis of SN-tSNE vs. t-SNE loss distributions using SW and KS statistical metrics.

Datasets	Algorithm	SW	KS
F-MNIST	t-SNE	0.7015	0.7163
	SN-tSNE	**0.9446**	**0.9272**
MNIST	t-SNE	0.2316	0.7891
	SN-tSNE	**0.6462**	**0.9545**
Olivetti	t-SNE	0.4324	0.9567
	SN-tSNE	**0.8649**	**0.9890**
USPS	t-SNE	0.8264	0.9911
	SN-tSNE	**0.9092**	**0.9918**
Pendigits	t-SNE	3.0×10^{-6}	0.0618
	SN-tSNE	**0.5517**	**0.7505**
COIL-20	t-SNE	0.0258	0.1763
	SN-tSNE	**0.8541**	**0.9417**
Isolet	t-SNE	0.3946	0.7320
	SN-tSNE	**0.6653**	**0.9071**
UCI-HAR	t-SNE	0.0154	0.2053
	SN-tSNE	**0.2327**	**0.6561**

computational efficiency, performs better than traditional t-SNE but still lags behind SN-tSNE, especially for datasets with more than 8000 samples. NMF is faster due to its efficient optimization process, relying on iterative multiplicative updates to factorize matrices, which avoids complex distance computations and large-scale pairwise operations. This leads to faster convergence, even for large datasets.

5.6. Shrinkage effect of MUR on inter-point distances

To empirically validate the shrinkage effect induced by the proposed MUR in SN-tSNE, we designed an experiment to monitor the evolution of average inter-point distances across iterations. Specifically, we computed the mean pairwise Euclidean distance among embedded points at each iteration during the training of both t-SNE and SN-tSNE. We applied this evaluation to two datasets: Pendigits and Isolet, chosen to assess the behavior of SN-tSNE in a diverse setting beyond image and non-image datasets. The implementations of both t-SNE and SN-tSNE are updated to record the low-dimensional embeddings at each iteration. At every step, we computed the average pairwise Euclidean distance between all embedded samples. This yields a shrinkage curve that reveals how the embedding space expands or contracts over time. The results are visualized in Fig. 7, where we observe a marked contrast between the behaviors of t-SNE and SN-tSNE:

- For t-SNE, the average pairwise distance increases monotonically, indicating a continuous expansion of the embedding space. This behavior reflects its aggressive minimization of divergence at the cost of compactness.
- In contrast, SN-tSNE maintains a tightly bounded distance range, confirming the shrinkage effect driven by the MUR. Even after 200+ iterations, SN-tSNE preserves a relatively compact embedding, suggesting enhanced regularization and cluster cohesion.

This behavior is particularly important for applications requiring stable and compact embeddings, such as clustering or interpretability-sensitive visualizations. Moreover, these results empirically confirm that the semi-non-negative formulation of SN-tSNE introduces geometric regularization without sacrificing the representation structure.

Table 6
Execution time comparison of SN-tSNE and other baselines in seconds.

	Dataset							
Algorithm	Pendigits	USPS	MNIST	FMNIST	COIL-20	Olivetti	Isolet	UCI-HAR
NMF	0.59	3.22	108.71	105.71	12.65	2.39	2.54	2.68
UMAP	228.62	1056.89	1073.22	1102.34	211.72	118.23	843.29	1013.66
t-SNE	59.29	1418.34	2585.53	2769.83	63.34	3.44	966.15	1079.62
PGD-tSNE	98.24	1921.73	2718.34	2785.62	84.98	54.95	1644.86	1704.15
MU-tSNE	1.31	726.18	1956.00	1934.20	85.19	7.55	625.85	849.26
SN-tSNE	8.48	623.09	748.84	776.09	14.7	3.07	365.00	581.81

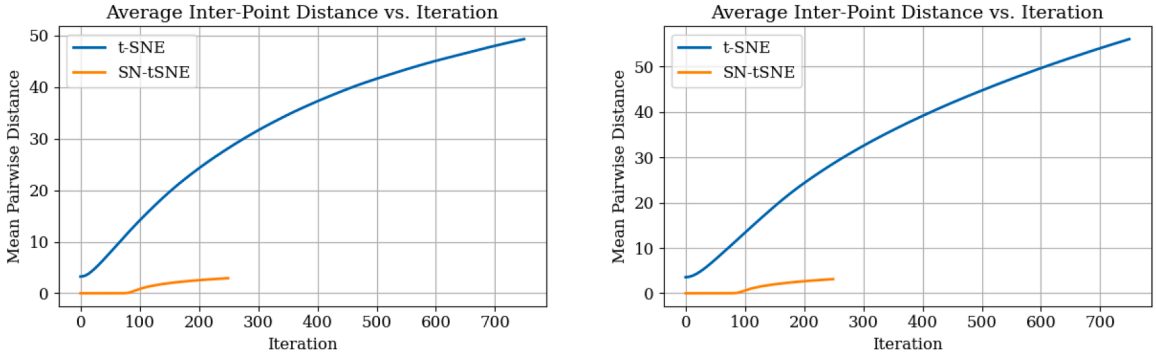


Fig. 7. Average inter-point distance vs. iteration for t-SNE and SN-tSNE on the Pendigits (left) and Isolet (right) datasets. The shrinkage effect of SN-tSNE is consistently observed, where it exhibits a bounded and contracting behavior.

5.7. Structure preservation evaluation

To assess the capability of SN-tSNE in preserving both local and global structures of high-dimensional data, we conduct a quantitative evaluation using three established metrics: GQ, MRRE, and Trustworthiness at large neighborhood size ($k = 100$). These metrics offer complementary insights:

- **Global Quality (GQ)** [37] quantifies the mean absolute distortion in pairwise distances between high- and low-dimensional spaces (lower is better).
- **Mean Relative Rank Error (MRRE)** [36] evaluates the preservation of global rank-ordering of neighbors across spaces (lower is better).
- **Trustworthiness** [39] measures the fidelity of preserving true high-dimensional neighbors in the low-dimensional embedding (higher is better).

Table 7 presents the results on eight benchmark datasets: FMNIST, MNIST, Olivetti, USPS, Pendigits, COIL-20, Isolet, and UCI-HAR, comparing UMAP, t-SNE, and SN-tSNE. Across all datasets, SN-tSNE consistently achieves the best MRRE scores, indicating superior preservation of global neighbor rankings. It also yields the highest trustworthiness values, confirming its ability to maintain meaningful neighborhood structures. While t-SNE generally outperforms UMAP in GQ, SN-tSNE further reduces this error in all datasets, e.g., FMNIST (0.130 vs. 0.149), MNIST (0.232 vs. 0.241), and Olivetti (0.164 vs. 0.173). Notably, on more complex datasets such as COIL-20 and Olivetti, SN-tSNE achieves a balance between global structure preservation (lower GQ and MRRE) and local neighborhood fidelity (higher trustworthiness), suggesting that the semi-non-negativity constraint and the multiplicative update mechanism enhance both interpretability and stability of the embedding. These findings quantitatively validate the structure-preserving capability of SN-tSNE and strengthen the visual patterns previously observed in the qualitative evaluations (see Section 5.9).

Table 7
Comparison of algorithms' performance in terms of structure preservation.

Dataset	Algorithm	GQ	MRRE	Trustworthiness
FMNIST	UMAP	0.202	0.0004	0.974
	t-SNE	0.149	0.0004	0.976
	SN-tSNE	0.130	0.0002	0.979
MNIST	UMAP	0.277	0.0005	0.947
	t-SNE	0.241	0.0005	0.950
	SN-tSNE	0.232	0.0004	0.953
Olivetti	UMAP	0.212	0.0135	0.841
	t-SNE	0.173	0.0122	0.854
	SN-tSNE	0.164	0.0113	0.870
USPS	UMAP	0.264	0.0005	0.954
	t-SNE	0.234	0.0005	0.959
	SN-tSNE	0.223	0.0003	0.965
Pendigits	UMAP	0.264	0.0031	0.927
	t-SNE	0.294	0.0024	0.934
	SN-tSNE	0.244	0.0013	0.936
COIL-20	UMAP	0.287	0.0032	0.855
	t-SNE	0.237	0.0021	0.922
	SN-tSNE	0.220	0.0019	0.931
Isolet	UMAP	0.196	0.0005	0.976
	t-SNE	0.181	0.0005	0.977
	SN-tSNE	0.178	0.0004	0.979
UCI-HAR	UMAP	0.161	0.0003	0.955
	t-SNE	0.135	0.0003	0.965
	SN-tSNE	0.123	0.0002	0.968

5.8. Objective evaluation

Settings. The NMF algorithm employs default settings defined in the Scikit-learn Python library, with the number of iterations set to 200. We utilized the umap-learn Python library[2] for UMAP, without leveraging Numba,[3] which is typically used to accelerate the UMAP algorithm. This choice was made to ensure a fair comparison with other algorithms. We maintained default parameters for UMAP, except for the number of neighbors, which was optimized for each dataset to achieve the best performance. It is set to 5, 5, 5, 10, 15, 15, 25, and 25 for Pendigit, Olivetti, Isolet, USPS, COIL-20, UCI-HAR, MNIST, and Fashion-MNIST, respectively. The parameter settings of t-SNE, PGD-tSNE, MU-tSNE, and SN-tSNE are performed based on Section 5.2. The configuration for t-SNE is chosen to be optimal, with (perplexity, number of iterations) = (5, 250), (5, 500), (10, 500), (20, 500), (20, 500), (25, 500), (35, 1000), and (35, 1000) for Pendigits, Isolet, Olivetti, COIL-10, UCI-HAR, USPS, MNIST, and F-MNIST, respectively. The PDG-tSNE and MU-tSNE are configured the same way as the t-SNE. Finally, SN-tSNE is configured as follows: (perplexity, number of iterations) = (5, 50), (5, 100), (5, 150), (10, 100), (15, 100), (15, 150), (20, 250), (20, 250) for Pendigits, Isolet, Olivetti, USPS, UCI-HAR, COIL-20, MNIST, and F-MNIST, respectively. Since all the algorithms are stochastic and not deterministic, we ran each of them on each dataset 10 times to compute the average performance and the standard deviation in terms of execution time and evaluation metrics.

Clustering Task. SN-tSNE is systematically evaluated against the original t-SNE and other dimensionality reduction techniques, using clustering performance on their respective embedded data representations

[2] https://umap-learn.readthedocs.io/en/latest/
[3] https://numba.pydata.org/

Table 8
Comparison of algorithms' performance in terms of clustering quality.

Dataset	Algorithm	Acc	NMI	SC	DBI
FMNIST	NMF	0.53 ± 0.00	0.52 ± 0.00	0.30 ± 0.00	1.13 ± 0.00
	UMAP	0.54 ± 0.02	0.61 ± 0.01	0.51 ± 0.01	0.68 ± 0.03
	t-SNE	0.57 ± 0.01	0.56 ± 0.01	0.43 ± 0.01	0.77 ± 0.01
	PGD-tSNE	0.53 ± 0.01	0.53 ± 0.01	0.39 ± 0.01	0.78 ±0.01
	MU-tSNE	0.44 ± 0.01	0.41 ± 0.01	0.40 ± 0.01	0.77 ± 0.01
	SN-tSNE	**0.58 ± 0.01**	**0.62 ± 0.01**	**0.54 ± 0.01**	**0.65 ± 0.02**
MNIST	NMF	0.55 ± 0.00	0.45 ± 0.00	0.19 ± 6.45	1.62 ± 0.00
	UMAP	0.80 ± 0.01	0.80 ± 0.01	0.58 ± 0.01	0.58 ± 0.03
	t-SNE	0.82 ± 0.02	0.77 ± 0.02	0.47 ± 0.01	0.71 ± 0.03
	PGD-tSNE	0.77±0.03	0.72 ± 0.01	0.41 ± 0.01	0.79 ± 0.02
	MU-tSNE	0.40 ± 0.01	0.37 ± 0.01	0.35 ± 0.01	0.82 ± 0.01
	SN-tSNE	**0.85 ± 0.02**	**0.82 ± 0.01**	**0.61 ± 0.01**	**0.54 ± 0.02**
Olivetti	NMF	0.52 ± 0.01	0.73 ± 0.01	0.22 ± 0.00	1.24 ± 0.01
	UMAP	0.65 ± 0.01	0.82 ± 0.01	0.69 ± 0.01	0.35 ± 0.01
	t-SNE	0.65 ± 0.01	0.82 ± 0.01	0.60 ± 0.02	0.50 ± 0.02
	PGD-tSNE	0.48 ± 0.03	0.72 ± 0.01	0.47 ± 0.01	0.56 ± 0.01
	MU-tSNE	0.31 ± 0.01	0.56 ± 0.02	0.37 ± 0.01	0.75 ± 0.01
	SN-tSNE	**0.67 ± 0.02**	**0.83 ± 0.01**	**0.74 ± 0.02**	**0.34 ± 0.02**
USPS	NMF	0.61 ± 0.00	0.58 ± 0.00	0.34 ± 0.00	1.26 ± 0.00
	UMAP	0.81 ± 0.02	0.85 ± 0.01	0.68 ± 0.01	0.46 ± 0.01
	t-SNE	0.80 ± 0.02	0.82 ± 0.02	0.45 ± 0.02	0.75 ± 0.02
	PGD-tSNE	0.69 ± 0.03	0.73 ± 0.01	0.40 ± 0.01	0.78 ± 0.01
	MU-tSNE	0.44 ± 0.01	0.44 ± 0.02	0.42 ± 0.01	0.80 ± 0.01
	SN-tSNE	**0.83 ± 0.02**	**0.87 ± 0.01**	**0.70 ± 0.02**	**0.43 ± 0.02**
Pendigits	NMF	0.63 ± 0.02	0.63 ± 0.02	0.26 ± 0.01	1.37 ± 0.01
	UMAP	0.89 ± 0.03	0.87 ± 0.02	0.67 ± 0.01	0.44 ± 0.02
	t-SNE	0.89 ± 0.04	0.86 ± 0.02	0.73 ± 0.01	0.41 ± 0.01
	PGD-tSNE	0.62 ± 0.02	0.65 ± 0.03	0.68 ± 0.02	0.41 ± 0.02
	MU-tSNE	0.51 ± 0.01	0.46 ± 0.02	0.38 ± 0.01	0.78 ± 0.01
	SN-tSNE	**0.92 ± 0.02**	**0.89 ± 0.02**	**0.75 ± 0.01**	**0.36 ± 0.01**
COIL-20	NMF	0.60 ± 0.01	0.71 ± 0.01	0.32 ± 0.01	1.41 ± 0.03
	UMAP	0.86 ± 0.02	0.91 ± 0.01	0.71 ± 0.01	0.51 ± 0.03
	t-SNE	0.83 ± 0.01	0.91 ± 0.01	0.67 ± 0.01	0.50 ± 0.01
	PGD-tSNE	0.71 ± 0.01	0.83 ± 0.01	0.57 ± 0.01	1.18 ±0.01
	MU-tSNE	0.52 ± 0.01	0.63 ± 0.01	0.46 ± 0.01	0.73 ± 0.01
	SN-tSNE	**0.88 ± 0.01**	**0.94 ± 0.01**	**0.77 ± 0.01**	**0.47 ± 0.02**
Isolet	NMF	0.40±0.01	0.54±0.02	0.20±0.01	1.40±0.01
	UMAP	0.58±0.02	0.77±0.02	0.61±0.03	0.55±0.04
	t-SNE	0.60±0.01	0.70±0.01	0.38±0.01	0.78±0.01
	PGD-tSNE	0.56±0.02	0.71±0.01	0.39±0.01	0.78±0.02
	MU-tSNE	0.36±0.01	0.51±0.01	0.33±0.02	0.83±0.04
	SN-tSNE	**0.63±0.01**	**0.79±0.01**	**0.69±0.01**	**0.43±0.01**
UCI-HAR	NMF	0.52±0.04	0.44±0.03	0.32±0.04	1.10±0.03
	UMAP	0.66±0.02	0.69±0.02	0.59±0.02	0.79±0.03
	t-SNE	0.65±0.01	0.57±0.01	0.31±0.02	0.94±0.03
	PGD-tSNE	0.57±0.02	0.50±0.02	0.38±0.03	0.79±0.02
	MU-tSNE	0.48±0.01	0.39±0.01	0.29±0.02	1.15±0.03
	SN-tSNE	**0.74±0.01**	**0.73±0.02**	**0.65±0.03**	**0.49±0.02**

as the primary evaluation criterion. The results, summarized in Table 8, consistently demonstrate that SN-tSNE outperforms all other techniques across all evaluation metrics and datasets. The significantly higher ACC and NMI scores achieved by SN-tSNE indicate its ability to preserve intrinsic data structures more effectively, leading to superior clustering performance. The SN-tSNE's DBI and SC scores confirm its ability to generate distinct and well-defined clusters, effectively mitigating the crowding issue inherent in t-SNE while surpassing UMAP in clustering quality. Thanks to the MUR that induces the non-negativity constraint with the shrinkage effect to improve convergence, and by promoting compact clusters and well-separated embedding spaces. This effect enhances the quality of the embedding by ensuring that similar points remain close while maintaining meaningful distances between dissimilar points. Importantly, the regularization induced by shrinkage prevents over-dispersion of the embedding, which could otherwise distort the local manifold structure. These properties collectively improve the stability and robustness of the optimization process, as evidenced by the superior clustering performance of SN-tSNE across all datasets, and achieve superior performance in dimensionality reduction and manifold embedding, establishing it as a robust and effective approach for such tasks.

In evaluating different techniques, NMF emerges as the fastest technique, albeit with the poorest performance among them. In the Pendigits dataset, UMAP and t-SNE exhibit competitive results in evaluation metrics. While UMAP performs better in external evaluation metrics, t-SNE achieves better results in terms of internal metrics. Evaluation metrics reveal the comparative inefficacy of PGD-tSNE, which is pitted against other techniques, underscoring the trade-offs between speed and performance. The same applies to MU-tSNE.

Classification Task. This section provides a classification-based comparative analysis with the same setting used in the previous experiment. Classification tasks were performed using the SVM Classifier on the embedding spaces generated by these methods, and their performance was assessed using the Accuracy metric. From the results reported in Table 9, SN-tSNE demonstrated significantly higher scores than the other embedding techniques in terms of all metrics. Our proposed method seems to better maintain the underlying structures and relationships within the data, resulting in a better classification.

5.9. Subjective evaluation (data visualization)

Figs. 8–11 represent a 2D visualization of F-MNIST, MNIST, Olivetti, USPS, Pendigits, COIL-20, Isolet, and UCI-HAR datasets using UMAP, t-SNE, PGD-tSNE, MU-tSNE, and the proposed SN-tSNE, respectively. NMF is excluded from this experiment because it is not mainly used for visualization. In this experiment, we discuss the compactness and separation criteria. Clear compactness and separation within and between clusters contribute to the interpretability of the visualization, as they allow for an intuitive understanding of the underlying data structure. The PGD-tSNE algorithm projects data points into a constrained region within the embedding feature space while also identifying and accounting for outliers. In contrast, SN-tSNE projects data points into distinct, well-separated clusters without producing outliers. Although t-SNE is known to preserve the local structure of the data, in many cases, it does not preserve the compactness of the clusters, as shown in Figs. 8–11. In the visualization case of the MNIST dataset, we can observe that t-SNE projected the cluster of the number '4' divided into two parts separated by the cluster of the number '9', which makes the intra-cluster relationships not preserved. Observing the visualization of MU-tSNE, it is very obvious how MU-tSNE struggles to separate clusters, making the data projection appear as a single block in the two-dimensional space. Regarding UMAP visualization of all datasets, we notice that it separates the clusters from each other by large distances in most cases, which prevents the intra-cluster relationships from being preserved and affects the global structure of the data. Furthermore, UMAP clips the data into dense clusters, which may seem visually appealing; however, this can inadvertently lead to a loss of discriminative power and hinder the interpretability and utility of the embedding for the clustering and classification tasks. In contrast, SN-tSNE can clearly separate clusters in the low-dimensional space and preserve the inter and intra-cluster relationships. The semi-non-negative constraint supported by the shrinkage effect induced by the MUR optimization plays an important role in understanding the spatial relationships between data points, making it easier to identify distinct groups or patterns within the data, as in the case of the mentioned example above, SN-tSNE separates very well the clusters of number '4' and '9' from each other. The SN-tSNE's compactness ensures that similar points are projected in the same cluster while balancing the separation of clusters by positioning dissimilar points farther apart. The F-MNIST dataset presents the greatest challenge for visualization across all algorithms. However, we can notice that SN-tSNE stands out by capturing finer details. For example, SN-tSNE could separate clusters of numbers '5', '6', and '9' from each other better than

Table 9
Comparison of algorithms' performance in terms of classification accuracy.

Algorithm	FMNIST	MNIST	Olivetti	USPS	Pendigits	COIL-20	Isolet	UCI-HAR
NMF	0.68± 0.01	0.87± 0.00	0.75± 0.00	0.86± 0.00	0.92± 0.00	0.88 ± 0.00	0.75± 0.01	0.86± 0.01
UMAP	0.71 ± 0.00	0.93 ± 0.00	0.87 ± 0.02	0.94 ± 0.00	0.95 ± 1.11	0.87± 0.02	0.66±0.01	0.87±0.02
t-SNE	0.71 ± 0.00	0.93 ± 0.00	0.90 ± 0.02	0.93 ± 0.00	0.92 ± 0.00	0.85 ± 0.04	0.73±0.01	0.87±0.01
PGD-tSNE	0.62± 0.01	0.90± 0.01	0.55± 0.04	0.91± 0.01	0.74± 0.01	0.63± 0.01	0.71± 0.01	0.85± 0.01
MU-tSNE	0.45± 0.00	0.46± 0.00	0.36± 0.00	0.57± 0.01	0.58± 0.02	0.66± 0.00	0.45± 0.01	0.56± 0.02
SN-tSNE	**0.72± 0.00**	**0.94± 0.00**	**0.95± 0.00**	**0.95± 0.00**	**0.96± 0.00**	**0.92± 0.00**	**0.76± 0.01**	**0.89± 0.01**

t-SNE and UMAP. Overall, UMAP is lauded for its ability to preserve both local and global structures, resulting in smoother, globally coherent embeddings. Conversely, t-SNE is known for its adeptness in capturing intricate local structures and emphasizing local clusters. Since the proposed method is based on t-SNE, it inherits these characteristics while potentially offering improved performance. The choice between all the methods depends on factors such as dataset characteristics, analysis goals, computational efficiency, and interpretability preferences.

While the above discussion is based on visual inspection, these qualitative observations are quantitatively supported by the clustering quality metrics reported earlier in Section 5.8. In particular, the SN-tSNE consistently achieves higher Silhouette scores, lower Davies-Bouldin Index values, and higher Calinski-Harabasz scores compared to baseline methods. These metrics confirm the improved intra-cluster compactness and inter-cluster separation observed visually in Figs. 8–11, thereby reinforcing the reliability of our subjective evaluation.

5.10. Interpretability analysis

To demonstrate the visual interpretability of the SN-tSNE algorithm compared to the t-SNE, we conduct an experiment that involves reconstructing images and analyzing the visual differences between the reconstructed images and the original images. Using the Olivetti face dataset and the approach of out-of-sample extension, we train a Multi-Layer Perceptron (MLP) regressor to map the low-dimensional embedding input data, produced by each algorithm, to the high-dimensional original data. This trained MLP applies the learned transformation to new, unseen embedded data. Thus, SN-tSNE-MLP refers to applying MLP to the embedded data generated by SN-tSNE to predict the original data, and the same applies to t-SNE-MLP. By computing the reconstruction error through the mean square error (MSE) between the predicted data (reconstructed images) and actual images, we compare the interpretability of the results produced by SN-tSNE vs t-SNE. Since NMF is renowned for its exceptional dimensionality reduction capabilities, while preserving data interpretability is included for comparison purposes. For NMF, the reconstruction of face image x_i^r is performed algebraically using the formula $x_i^r = W h_i$, where h_i is the low-rank embedding of the original face image x_i. Table 10 presents the reconstruction MSE for five random Olivetti samples, along with the average MSE across the entire Olivetti dataset. The experiment results reveal that SN-tSNE-MLP achieves the best reconstruction scores among all methods. Thanks to the MUR, SN-tSNE generates more negative points compared to t-SNE and promotes non-negativity in the embedded space, resulting in improved data reconstruction and approaching the performance of NMF in terms of reconstruction's MSE. Thus, while SN-tSNE is more powerful than NMF due to its t-SNE nature, it also becomes closer to NMF in terms of reconstruction. The best second scores are for NMF, which is expected, as NMF is fully non-negative and designed primarily for dimensionality reduction while preserving interpretability. Fig. 12 shows five random samples to visually analyze the difference between the outputs of the algorithms. For most of the face images, SN-tSNE-MLP leads to better visual reconstruction. The outputs of SN-tSNE-MLP are more interpretable compared to NMF and t-SNE-MLP, as evidenced by the clear reconstruction of facial features, particularly distinct parts such as mouths, eyes, and noses. In contrast, t-SNE-MLP generates noisy and blurred reconstructions with distorted. For example, in the first row, t-SNE-MLP generated noise around the mouth and nose, causing a loss of detail. NMF added artifact-laden facial features, such as the glasses in the face of the first-row sample, as it learned that glasses were a discriminative feature due to their presence in some other faces. In contrast, SN-tSNE-MLP preserved facial feature edges more accurately without introducing artifacts, resulting in better interpretability and competing with NMF, unlike t-SNE.

Table 10
Image reconstruction MSE of the samples shown in Fig. 12, along with the average reconstruction MSE for the Olivetti face dataset.

	NMF	t-SNE-MLP	SN-tSNE-MLP
Sample 1	0.0037	0.0119	0.0028
Sample 2	0.0047	0.0076	0.0020
Sample 3	0.0033	0.0053	0.0018
Sample 4	0.0023	0.0088	0.0010
Sample 5	0.0026	0.0049	0.0009
Olivetti MSE	0.0028	0.0085	0.0023

6. Discussion

This work investigated whether enforcing semi-non-negativity via a multiplicative update rule (MUR) can improve the interpretability and stability of manifold embeddings while preserving t-SNE's local-structure strengths. Across controlled synthetic data and diverse real-world datasets, SN-tSNE consistently yields tighter clusters, steadier optimization, and competitive, often improved, global structure as quantified by GQ, MRRE, and trustworthiness.

Answers to the research questions. *(RQ1: Does non-negativity improve interpretability and compactness?)* Yes. Higher Non-Negative Points (NNP) ratios correlate with better external clustering results (Accuracy and NMI) and improved internal clustering results (Silhouette and DBI), and removing negative-coordinate points further boosts clustering quality. *(RQ2: Does MUR induce beneficial shrinkage?)* Yes. The multiplicative dynamics reduce average inter-point distances within clusters and damp oscillations, leading to earlier stabilization and more compact embeddings. *(RQ3: Is global structure preserved?)* Yes. SN-tSNE maintains, and at times improves, global fidelity (lower GQ, lower MRRE, strong trustworthiness), indicating compaction without inter-cluster collapse.

Synthetic vs. real data. In controlled synthetic datasets, where ground-truth structure and separability are explicit, the NNP ratio correlates strongly with clustering scores, isolating the effect of semi-non-negativity from confounders. On real-world datasets, improvements persist but are moderated by intrinsic noise, overlapping classes, and high intra-class variability. This distinction clarifies the causal mechanism, semi-non-negativity leads to compactness, and this means better clustering, and underscores robustness across both controlled and practical scenarios, including all used datasets.

Positioning relative to prior work. Compared to classical t-SNE's additive updates, SN-tSNE's multiplicative rule introduces a shrinkage bias that curbs negative excursions and stabilizes trajectories while

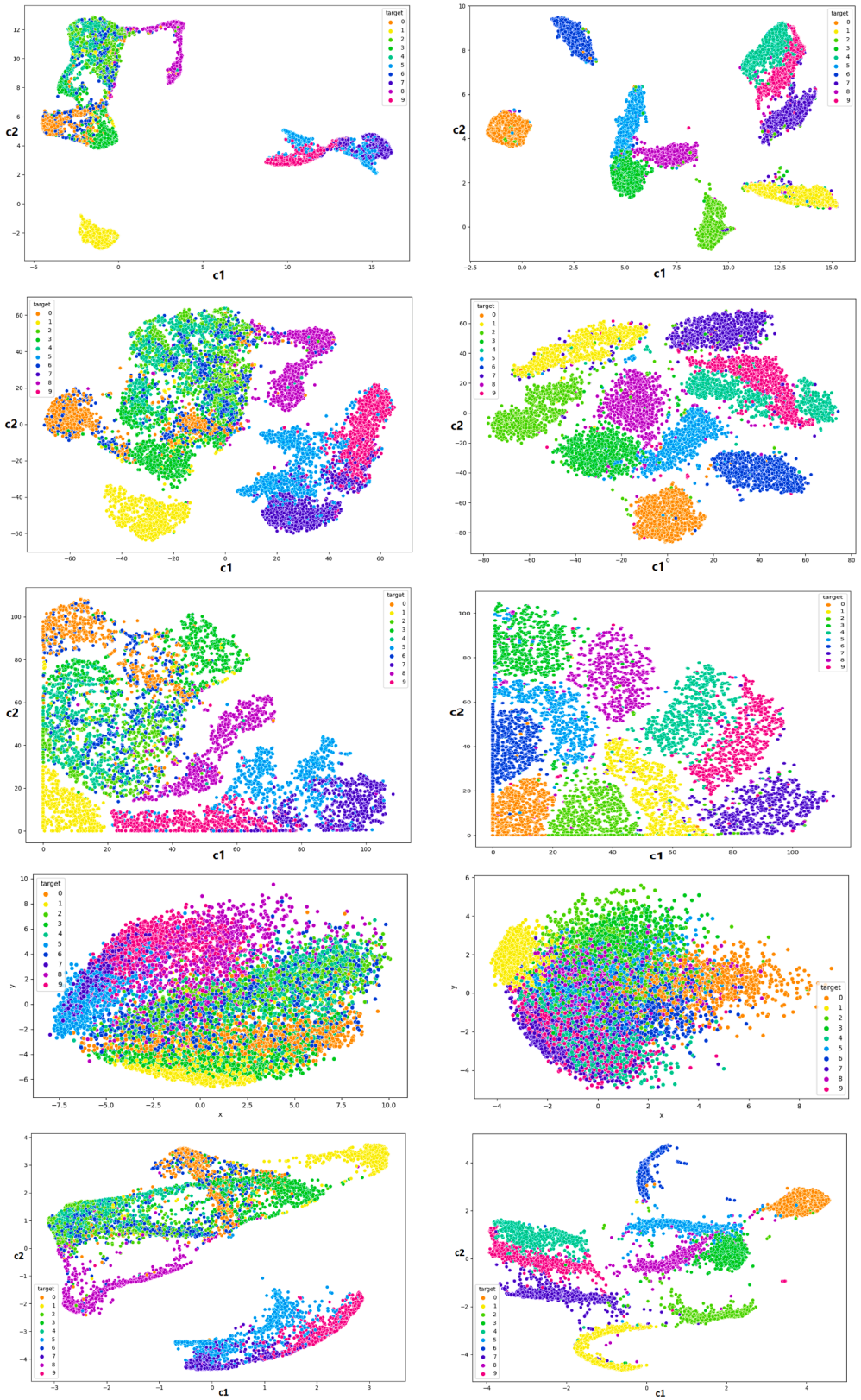


Fig. 8. Data visualization: from left to right, F-MNIST and MNIST datasets using, from top to down, UMAP, t-SNE, PGD-tSNE, MU-tSNE, and SN-tSNE embedding techniques.

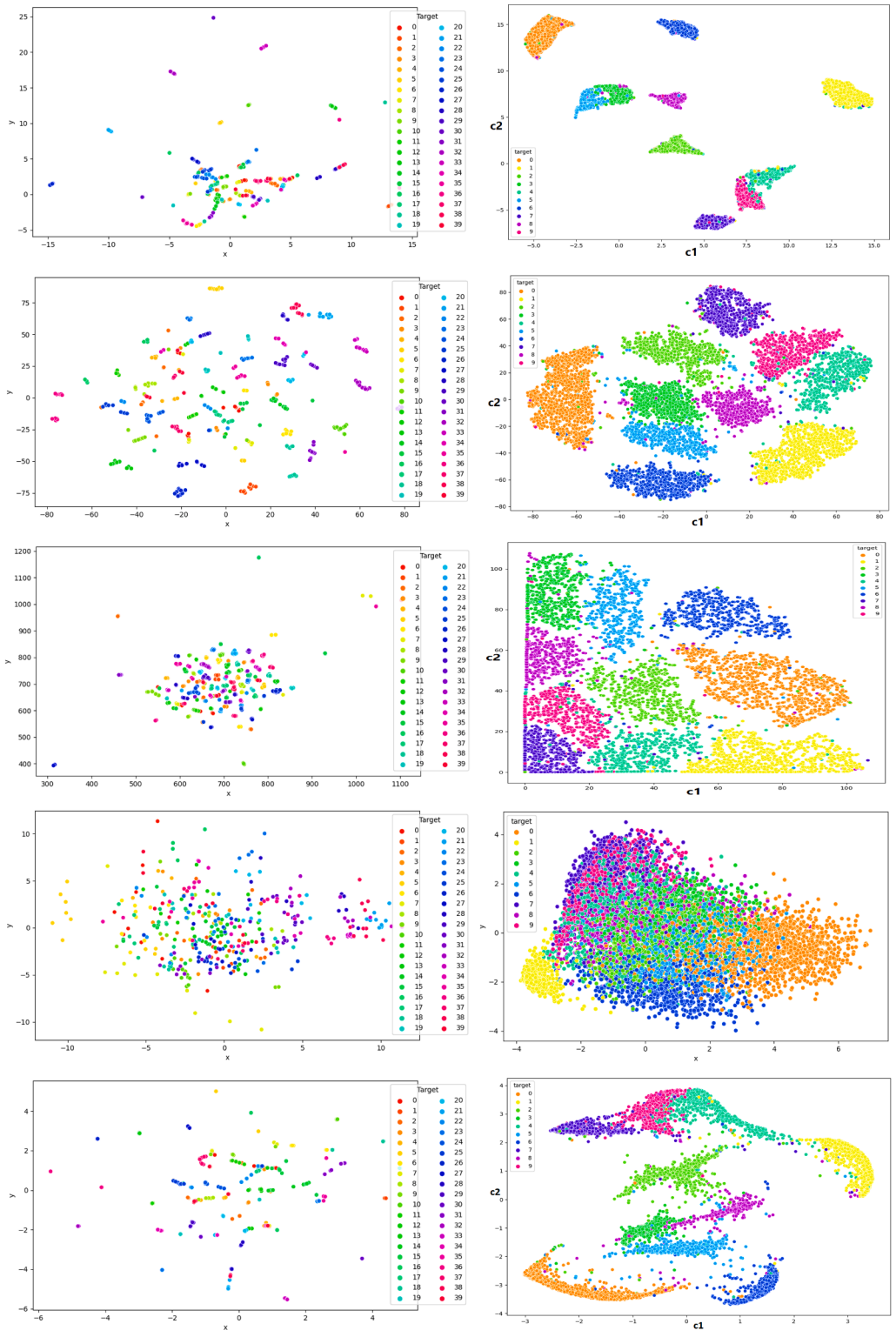


Fig. 9. Data visualization: from left to right, Olivetti and USPS datasets using, from top to down, UMAP, t-SNE, PGD-tSNE, MU-tSNE, and SN-tSNE embedding techniques.

preserving neighborhood fidelity. Relative to UMAP, SN-tSNE prioritizes compact, interpretable clusters with strong local metrics; while UMAP remains attractive for very large-scale speed, SN-tSNE narrows the gap by stabilizing earlier. Unlike full non-negativity schemes (e.g., NMF), our semi-non-negative formulation avoids redefining the objective, retaining t-SNE's characteristic manifold preservation.

Practical guidance. SN-tSNE is most advantageous when (i) coordinate interpretability matters, (ii) cluster compactness and neighborhood fidelity are primary goals, and (iii) moderate sample sizes allow faithful affinity estimation. Default settings similar to t-SNE work well; multiplicative updates tend to require fewer effective iterations due to earlier stabilization.

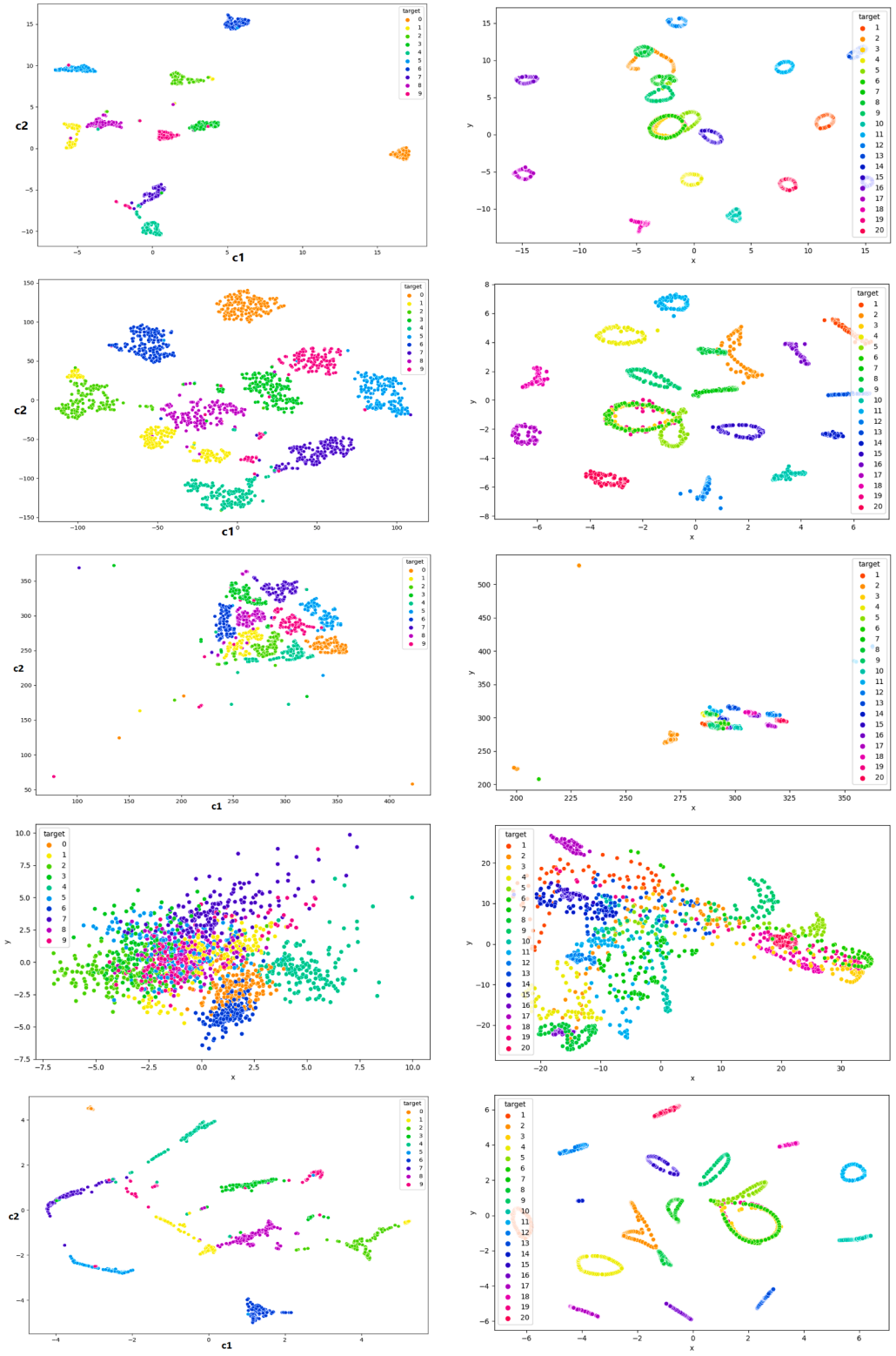


Fig. 10. Data visualization: from left to right, Pendigits and COIL-20 datasets using, from top to down, UMAP, t-SNE, PGD-tSNE, MU-tSNE, and SN-tSNE embedding techniques.

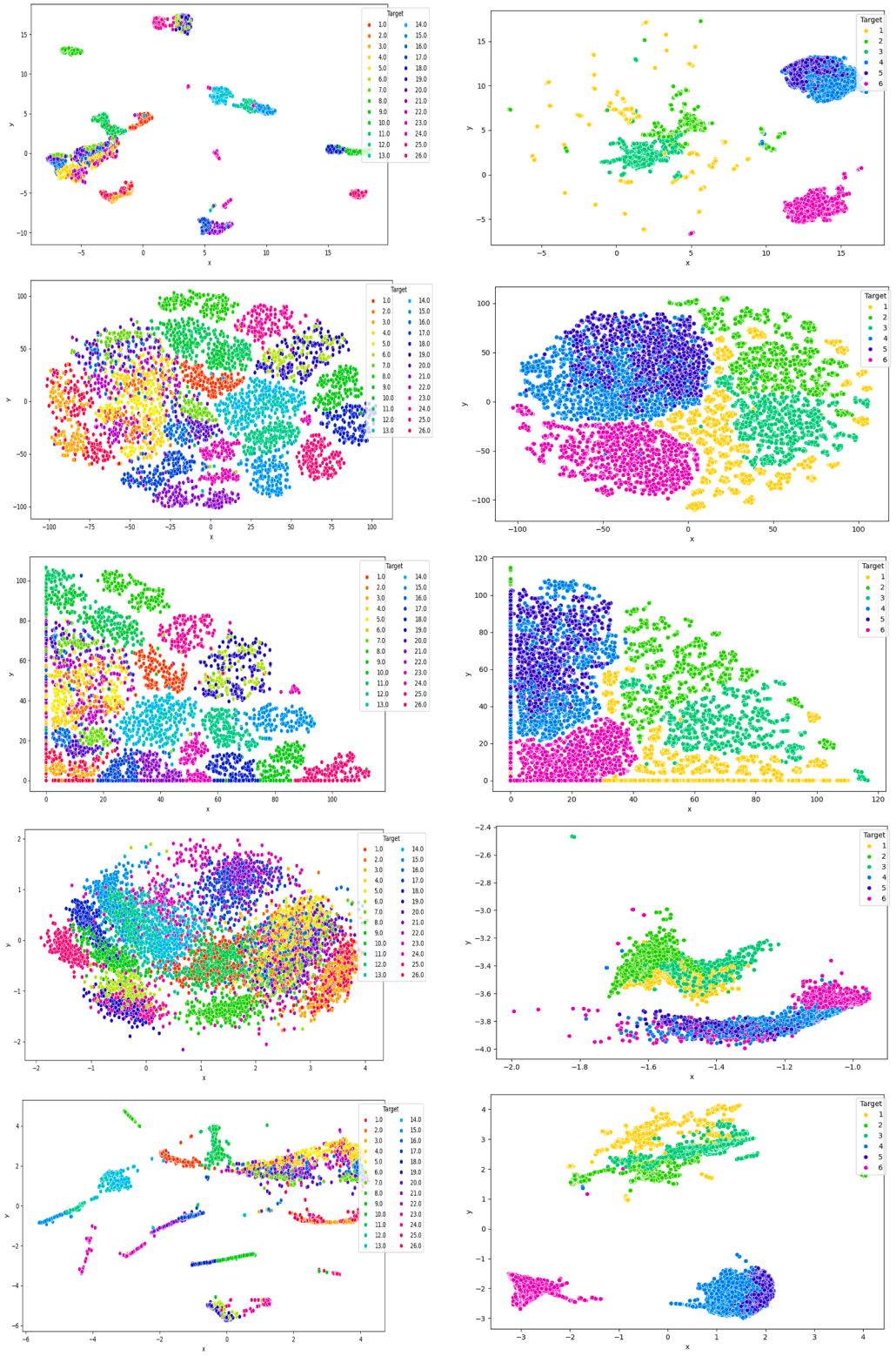


Fig. 11. Data visualization: from left to right, Isolet and UCI-HAR datasets using, from top to down, UMAP, t-SNE, PGD-tSNE, MU-tSNE, and SN-tSNE embedding techniques.

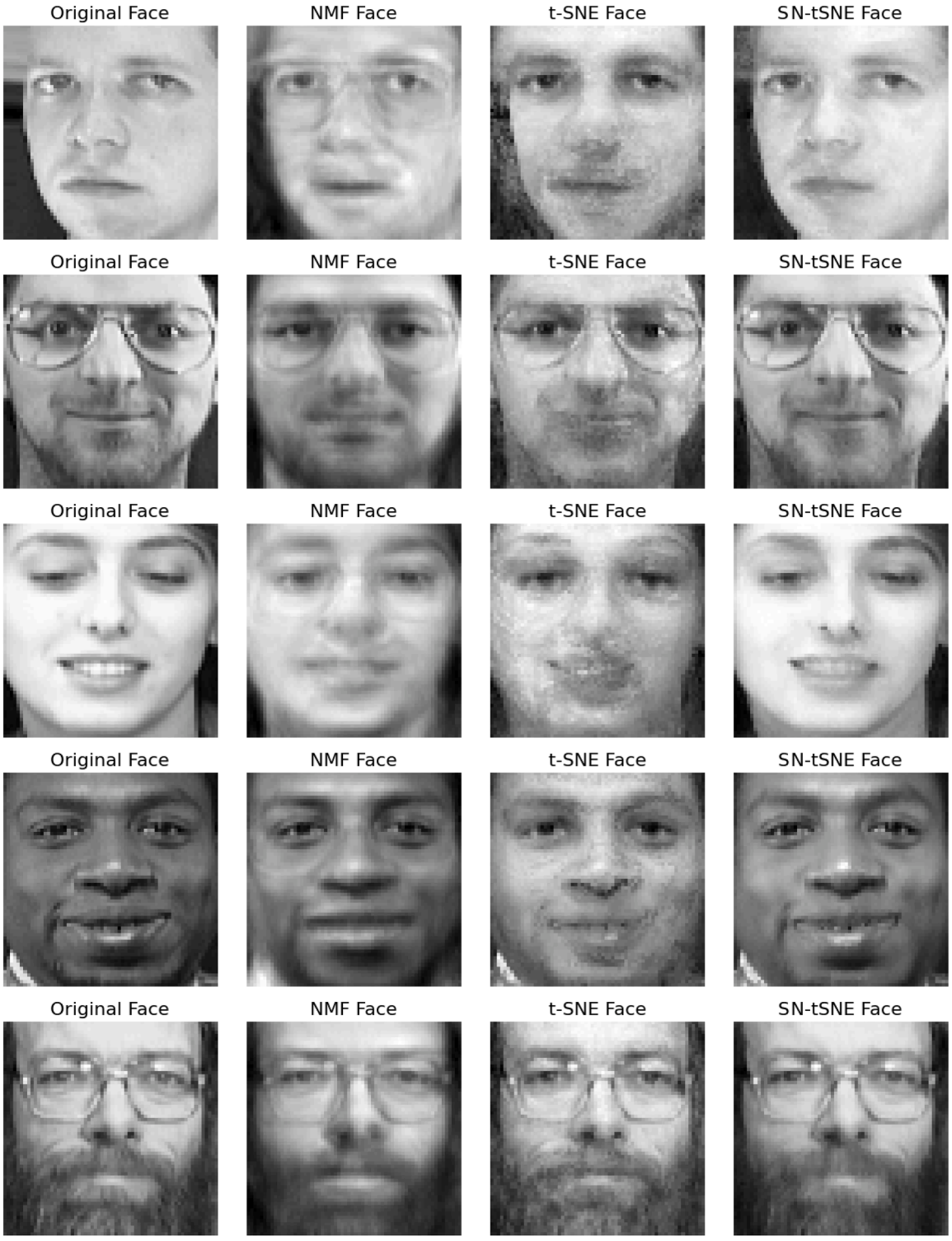


Fig. 12. Visualization of Olivetti face samples reconstructed from NMF, t-SNE, and SN-tSNE embeddings.

Limitations and failure modes. Datasets whose semantics rely on signed relations may benefit less from non-negativity. Under extreme class overlap or heavy noise, gains are naturally attenuated. For very large N under strict runtime budgets, fast approximations (e.g., UMAP) can remain attractive, though SN-tSNE's stability reduces tuning burden. As with t-SNE, initialization and perplexity can influence fine structure; we mitigate this with multiple runs and metric triangulation.

Convergence perspective. The semi-non-negative, multiplicative update alters the optimization landscape by (i) biasing updates to remain in interpretable regions when initialized non-negative, and (ii) introducing a quadratic shrinkage term that contracts intra-cluster distances while preserving inter-cluster separation. Empirically, this yields fewer oscillations and earlier stabilization, consistent with the analysis in Section 4.2.

7. Conclusion

This paper introduces SN-tSNE, an enhanced version of t-SNE with a novel MUR, transforming the traditional additive gradient descent approach. This innovation enforces non-negativity in the embedded feature space, improving interpretability and visualization while accelerating convergence by requiring fewer iterations. SN-tSNE demonstrates significant advancements over conventional t-SNE and similar techniques, producing embeddings better suited for clustering and classification tasks and intuitive visualizations. Experimental results on diverse datasets validate its superior performance in capturing underlying data structures. Despite the improvements of SN-tSNE, future work will focus on achieving full non-negativity in t-SNE embeddings to further enhance clarity and applicability. Additionally, we plan to analyze the relationship between perplexity and cluster structure using synthetic data and extend non-negativity enforcement to other techniques like UMAP.

CRediT authorship contribution statement

Mebarka Allaoui: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization; **Rachid Hedjam:** Writing – review & editing, Validation, Methodology, Formal analysis, Conceptualization; **Khadra Bouanane:** Writing – review & editing, Formal analysis; **Mohand Saïd Allili:** Writing – review & editing, Formal analysis; **Mohammed Lamine Kherfi:** Writing – review & editing, Formal analysis; **Samir Brahim Belhaouari:** Writing – review & editing, Formal analysis.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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